

Macroeconomic Sentiments and Job Search Behavior

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Abstract

Employed workers who expect the labor market to worsen perceive a greater risk of losing their own jobs and search more intensively while employed. Using survey data, I show that this relationship is driven by workers' fear of job loss rather than their expectations about finding a new job. I interpret this as precautionary on-the-job search: workers look harder for outside offers to protect themselves against the possibility of unemployment. To quantify the aggregate implications, I embed this mechanism in a search-and-matching model with cyclical separation risk and on-the-job search. In the calibrated model, workers search 27 percent more in recessions than in expansions. This precautionary behavior means that fewer workers who lose their jobs end up unemployed, reducing the recessionary rise in unemployment by about 16 percent. The reason is simple: outside offers matter because they let some separated workers move directly to a new job rather than enter unemployment.

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1 Introduction

Do households adjust their job search effort before labor market conditions worsen? Recessions make job search significantly harder due to the scarcity of jobs, an increasing pool of job seekers and the increased risk of unemployment. If households foresee an economic downturn in the future, they may also perceive a higher probability of losing their own job and to search while still employed to mitigate this risk. Using individual-level survey data from the United States, I show that employed workers who are pessimistic about the economy also expect higher risk of job-loss and search more on the job. I then use these facts to discipline a search-and-matching model and quantify their implications for unemployment dynamics.

This paper makes three contributions. First, I show that pessimism about the economy and perceived job-loss risk strongly predicts search intensity among employed workers. Second, I show that the response is consistent with precautionary search. It is tied to perceived job-loss risk and job-security motives. Third, I quantify the aggregate implications in a Diamond–Mortensen–Pissarides model with endogenous on-the-job search ([Mortensen and Pissarides, 1994](#)). In the model, workers search more in recessions because job-loss risk is higher, and this response reduces unemployment inflows. Workers who search more are more likely to have an outside offer when job loss occurs, and those offers help them avoid unemployment. When perceived job-loss risk rises, the value of this option rises as well. Search therefore affects aggregate flows by changing how many separations become unemployment spells. In the calibrated model, this margin reduces the recessionary rise in unemployment by about 16 percent.

The empirical analysis uses the Survey of Consumer Expectations (SCE), a monthly panel that elicits probabilistic expectations alongside detailed measures of on-the-job search ([Armantier et al., 2017](#)). A one-standard-deviation increase in macro pessimism, measured as the subjective probability that unemployment will be higher in 12 months, is associated with roughly 24 additional minutes of search per week among employed workers. This association operates largely through personal labor-market expectations. Workers who are more pessimistic about aggregate unemployment report higher job-loss risk, while their expected job-finding probability conditional on job loss moves much less. Macro pessimism predicts search mainly when it is reflected in workers' own perceived risk of job loss.

Three pieces of evidence point to a precautionary interpretation. Among employed workers who search, higher perceived job-loss risk predicts a higher probability of citing job security as a search motive. The search response is also strongest among workers who

report both high job-loss risk and weak re-employment prospects, the case in which an outside offer is most useful. Finally, perceived job-finding probability enters negatively: workers who expect to find work quickly if separated search *less*. This pattern is hard to reconcile with a pure job-ladder story, but it follows naturally if search partly insures against unemployment.

Reported job-loss probabilities are also predictive of later separations. Workers who assign a higher probability to losing their job are more likely to separate over the subsequent three months, and the relationship is monotone across belief quintiles. This is consistent with the idea that some workers have advance information about job loss (Hendren, 2017).¹ The same beliefs that predict search also forecast later separations. I use these facts to motivate and discipline the model.

To study aggregate implications, I develop a Diamond–Mortensen–Pissarides model with two aggregate states, expansions and recessions. Employed workers face state-dependent separation risk and choose search intensity. Outside offers arrive with a probability that depends on search effort and labor-market tightness. If a worker holds an offer when separated, she moves directly into a new job rather than unemployment. The search first-order condition delivers the key comparative static: higher perceived job-loss risk raises the marginal benefit of search.

The counterfactuals quantify this mechanism. In the calibrated model, workers search about 27 percent more in recessions than in expansions. When separation risk is held constant, search becomes *procyclical* because lower market tightness discourages effort. Holding search fixed at its expansion level raises recession unemployment by about 0.6 percentage points. Thus precautionary search reduces the recessionary rise in unemployment by about 16 percent. The effect comes from a straightforward margin: workers who search more are more likely to have an offer when job loss occurs, and those offers help them avoid unemployment.

I also isolate the belief channel in a separate exercise. Starting from the recession state, I raise perceived job-loss risk by 10 percentage points while holding the actual separation risk and the aggregate state fixed. This is not a recession shock. It asks what happens when workers feel more exposed to job loss even though the objective labor market is unchanged. Workers respond by searching more, which raises the probability that they hold an outside offer. When offers can help workers avoid unemployment, the employment-to-unemployment probability falls and unemployment declines slightly.

¹Advance information and precautionary behavior are distinct. I show that workers who perceive higher risk adjust search in a way that is consistent with self-insurance, especially when they are pessimistic about re-employment.

When I shut down that insurance role of offers, the same belief shock has essentially no effect on unemployment. The exercise shows that beliefs matter for unemployment in the model because they change search before job loss, and because outside offers can turn some potential unemployment spells into direct job-to-job moves.

The paper relates to three strands of work. First, it contributes to research on subjective expectations and labor-market behavior. Workers' beliefs about labor-market prospects are informative and behaviorally relevant: employed workers have advance information about job loss ([Hendren, 2017](#)), and unemployed job seekers' beliefs predict outcomes but can be biased ([Spinnewijn, 2015](#); [Mueller et al., 2021](#)). I add evidence on a different margin: employed workers' perceived job-loss risk predicts how much they search before any separation occurs.

Second, the paper relates to the on-the-job search and job-ladder literature. Standard models emphasize search for better jobs and higher wages ([Burdett and Mortensen, 1998](#); [Postel-Vinay and Robin, 2002](#); [Menzio and Shi, 2011](#); [Moscarini and Postel-Vinay, 2018](#)). Recent work shows that employed search is quantitatively important and often cyclical. [Faberman et al. \(2022\)](#) document that on-the-job search is pervasive and effective, and [Ahn and Shao \(2017\)](#) show that employed search is countercyclical and related to precautionary concerns. [Simmons \(2024\)](#) studies precautionary on-the-job search before job loss and its implications for mobility and insurance. Relative to these papers, I use direct subjective job-loss probabilities to separate perceived separation risk from perceived job-finding prospects, use stated search motives to discipline the interpretation, and quantify the unemployment-inflow implications.

Third, the paper contributes to work on expectations and business-cycle fluctuations. Survey expectations matter for household decisions, and disagreement in macro beliefs is widespread ([Mankiw et al., 2003](#); [Coibion et al., 2018](#)). In labor markets, macro perceptions can affect personal expectations and behavior ([Roth and Wohlfart, 2019](#); [Carroll and Dunn, 1997](#)). [Pilossoff and Ryngaert \(2024\)](#) show that inflation expectations affect employed search through nominal wages. Here, perceived job-loss risk changes search because outside offers help workers avoid unemployment. This also connects the paper to work on search effort as a stabilizing force. [Mukoyama et al. \(2018\)](#) show that countercyclical search among the unemployed dampens unemployment fluctuations; this paper studies a related inflow margin among employed workers.

The rest of the paper is organized as follows. Section 2 documents the empirical relationship between perceived separation risk and on-the-job search. Section 3 develops a search model with precautionary on-the-job search and derives analytical results. Section 4 calibrates the model and reports untargeted model implications. Section 5 quantifies the

aggregate implications of precautionary search. Section 6 concludes.

2 Perceived Separation Risk and On-the-Job Search

This section uses individual-level data from the Survey of Consumer Expectations (SCE) to study how employed workers' labor-market expectations relate to on-the-job search. Workers who report higher job-loss risk search more while employed. Several results point to a precautionary motive: reported job-loss risk predicts subsequent separations, predicts job-security search motives, and matters most when perceived re-employment prospects are weak. I interpret the estimates as descriptive. They compare workers observed in similar time and location environments, but they do not isolate exogenous variation in beliefs.

The analysis proceeds in four steps. Section 2.2 relates expectations to search. Section 2.3 shows that macroeconomic pessimism is more closely related to personal job-loss expectations than to perceived job-finding prospects. Section 2.4 shows that reported job-loss probabilities forecast realized separations. Section 2.5 uses stated search motives and interaction patterns to distinguish precautionary search from other reasons for on-the-job search.

2.1 The Survey of Consumer Expectations

I use the Survey of Consumer Expectations (SCE), a nationally representative monthly internet-based panel conducted by the Federal Reserve Bank of New York since 2013. Each month approximately 1,300 household heads complete the survey; respondents typically remain in the panel for 12 consecutive months. The SCE has two components relevant to this paper. The *core monthly survey* collects employment status and probabilistic expectations—elicited on a 0–100 scale—about aggregate and personal labor market outcomes. The *Labor Market Module* (LMM) is administered to the same respondents every four months, so each participant completes it up to three times during their panel tenure. The LMM collects detailed information on job search behavior.

Appendix Figure A1 plots the time series of average macro unemployment expectations and average search hours among employed workers. Expectations vary substantially over the sample period, particularly during the COVID-19 recession. Search hours are volatile, reflecting the smaller rotating sample in the LMM, and do not display a mechanically smooth cyclical pattern.

Labor Market Module. The LMM’s primary search measure asks: “How many hours did you spend in the past 7 days looking or applying for jobs?” This question captures the *intensive margin* of on-the-job search. The module also records which search activities the respondent undertook (browsing postings, applying online, networking, contacting employers directly) and, for the mechanism tests in Section 2.5, the respondent’s stated reasons for searching. The analysis sample consists of 4,226 employed worker-months with non-missing search hours, spanning March 2014 through December 2023.²

Expectation variables. The empirical analysis relies on four probabilistic expectation variables from the SCE core survey, each elicited on a 0–100 scale as a point estimate rather than a full density forecast. *Macro pessimism* is the respondent’s subjective probability that “12 months from now the unemployment rate will be higher than it is now” (sample mean: 39.4%, SD: 23.9 pp). *Job-loss probability* is the percent chance the respondent assigns to involuntarily losing their current job in the next 12 months (mean: 21.3%, SD: 23.4 pp). *Voluntary separation probability* is the percent chance of voluntarily leaving the current job in the next 12 months (mean: 40.2%, SD: 30.4 pp). *Job-finding probability conditional on job loss* is the percent chance of finding a new job within three months if the current job is lost (mean: 57.1%, SD: 30.0 pp). Expected inflation is also collected in the core survey and enters the regressions as a control. Because the survey elicits point estimates, workers with the same reported probability may differ in subjective uncertainty; this heterogeneity is not observed.

Appendix Table A1 reports summary statistics. The average employed worker in the LMM reports approximately 3.8 hours per week of search (SD: 4.8 hours).

2.2 Perceived risk and search intensity

This subsection relates expectations to search intensity. I condition on a rich set of controls and fixed effects, so the estimates compare workers facing the same time and location environment.

The baseline specification regresses weekly search hours on the worker’s subjective probability that unemployment will be higher in 12 months:

$$\text{SearchHours}_{ist} = \beta E_{it} [\Delta \Pr(U_{t+12} \geq 0)] + \mathbf{X}'_{it} \gamma + \mu_s + \lambda_t + \varepsilon_{ist}, \quad (1)$$

²The full SCE Labor Market Module contains approximately 12,000 respondent-wave observations over this period; the analysis sample restricts to employed workers with non-missing search hours and expectation variables.

where $E_{it}[\Delta \Pr(U_{t+12} \geq 0)]$ is scaled to $[0, 1]$, $\mathbf{X}_{it}$ includes demographic controls, and μ_s and λ_t denote state and time fixed effects. Standard errors are clustered at the individual level. Figure 1 (Panel A) displays the conditional relationship between macro pessimism and search hours after residualizing both variables with respect to controls and fixed effects.



Notes: Each panel plots binned means of weekly search hours against a single expectation measure after residualizing both variables on the full set of controls, state fixed effects, and time fixed effects. Expectations are scaled to $[0, 1]$ and search hours are winsorized at the 5th and 95th percentiles. Sample: employed respondents in the SCE Labor Market Module.

Figure 1. Binscatter: expectations vs. search hours

Column (1) of Table 1 reports the corresponding coefficient. A one-standard-deviation increase in macro pessimism (24 percentage points) is associated with about 24 additional minutes of job search per week, roughly 10 percent of mean search. The estimate changes little when I include expectations about other macro variables, including inflation, stock prices, credit access, house prices, interest rates, and government debt. Figure 1 (Panel A) shows the corresponding conditional binscatter.

Aggregate beliefs are informative, but workers also hold beliefs about their own labor market prospects. Columns (2)–(5) of Table 1 therefore incorporate personal expectations about job finding and separation. Two results stand out: perceived *job-loss probability* strongly predicts higher search intensity, while perceived *job-finding probability conditional on job loss* enters with the opposite sign. Workers who expect to find a job quickly if separated search *less*. These signs are consistent with precautionary search: an outside offer is more valuable when job-loss risk is high and unemployment would be costly. Voluntary separation expectations are included as an additional control and enter positively, consistent with a job-ladder margin;³ the central mechanism in this paper is the perceived job-loss channel.

After conditioning on personal expectations, the macro pessimism coefficient falls but remains positive. This decline suggests that macro pessimism predicts search in part

³The job-ladder literature emphasizes that employed workers search to climb toward better-paying jobs; see Postel-Vinay and Robin (2002) and Moscarini and Postel-Vinay (2018) for theory and evidence.

Table 1. Expectations and On-the-Job Search Intensity

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
$E_t(\Delta \Pr(U_{t+12} \geq 0))$	1.607*** (0.372)	0.953*** (0.356)	0.953*** (0.355)	0.872** (0.370)	0.928*** (0.352)
$E_t(\Pr(\text{Lose Job}_{t+12}))$		2.046*** (0.425)	1.836*** (0.429)	1.629*** (0.441)	2.682*** (0.831)
$E_t(\Pr(\text{Leave Job}_{t+12}))$		1.567*** (0.330)	1.884*** (0.333)	1.313*** (0.361)	1.890*** (0.334)
$E_t(\Pr(\text{Find Job}_{t+3}))$			-1.156*** (0.294)	-1.568*** (0.298)	-0.877** (0.365)
$E_t(\Pr(\text{Offer}_{t+3}))$				0.019*** (0.003)	
$E_t(\Pr(\text{Lose Job}_{t+12})) \times$ $E_t(\Pr(\text{Find Job}_{t+3}))$					-1.479 (1.214)
$E_t(\pi_{t+12})$	-0.025 (0.021)	-0.025 (0.022)	-0.028 (0.022)	-0.024 (0.023)	-0.029 (0.022)
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	4132	4131	4130	3922	4130
R-squared	0.125	0.148	0.152	0.163	0.153

Notes: Dependent variable is weekly job search hours (winsorized). Expectations are scaled to [0, 1]: macro unemployment is $\Delta \Pr(U_{t+12} \geq 0)$, job loss and leave are 12-month probabilities, job-finding is $\Pr(\text{find within 3 months} \mid \text{lose})$, and P(offer) is the 3-month offer probability. Sample: employed LMM respondents. All specifications include time and state fixed effects, are weighted by survey weights, and control for age polynomial, education, race, gender, marital status, and household income. Standard errors clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

because it is related to workers' own perceived job-loss risk. The remaining coefficient may capture other aggregate risks not covered by the SCE's personal expectation questions.

To make the comparison more discrete, Table 2 repeats the same exercise using indicators for whether each of the four core expectations lies above the sample median. This specification compares workers with relatively high beliefs to those at or below the median rather than tracing marginal responses to a one-unit change in reported probabilities.

The indicator specification delivers the same qualitative pattern. Workers with above-median macro pessimism search about 0.5–0.8 hours more per week, and workers with above-median perceived job-loss risk search about 0.5–0.9 hours more per week, depending on the specification. Above-median voluntary-separation beliefs are also associated with more search, while above-median job-finding beliefs are associated with less search in the

Table 2. Expectations and On-the-Job Search Intensity: Above-Median Indicators

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
Above median: $E_t(\Delta \Pr(U_{t+12} \geq 0))$	0.765*** (0.174)	0.519*** (0.165)	0.520*** (0.165)	0.538*** (0.170)	0.512*** (0.164)
Above median: $E_t(\Pr(\text{Lose Job}_{t+12}))$		0.673*** (0.170)	0.588*** (0.170)	0.496*** (0.174)	0.860*** (0.248)
Above median: $E_t(\Pr(\text{Leave Job}_{t+12}))$		0.977*** (0.176)	1.075*** (0.176)	0.751*** (0.193)	1.077*** (0.177)
Above median: $E_t(\Pr(\text{Find} \text{Lose}_{t+3}))$			-0.595*** (0.169)	-0.836*** (0.170)	-0.311 (0.229)
$E_t(\Pr(\text{Offer}_{t+3}))$				0.019*** (0.003)	
Above median: $E_t(\Pr(\text{Lose Job}_{t+12})) \times$ Above median: $E_t(\Pr(\text{Find} \text{Lose}_{t+3}))$ $E_t(\pi_{t+12})$	-0.023 (0.021)	-0.024 (0.022)	-0.027 (0.022)	-0.022 (0.023)	-0.604* (0.341)
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	4132	4131	4130	3922	4130
R-squared	0.125	0.142	0.145	0.158	0.146

Notes: Dependent variable is weekly job search hours (winsorized). The four core expectations from Table 1 are replaced by indicators for responses strictly above the sample median in the employed sample with non-missing search hours; coefficients therefore measure search-hour differences relative to workers at or below the median. Column (4) retains the continuous expected-offer control, and Column (5) interacts the above-median job-loss and job-finding indicators. All specifications include time and state fixed effects, are weighted by survey weights, and control for age polynomial, education, race, gender, marital status, and household income. Standard errors clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

additive specifications. The interaction column sharpens the precautionary interpretation. Among workers with below-median job-finding beliefs, moving above the median in perceived job-loss risk raises search by about 0.86 hours per week; when job-finding beliefs are also above the median, this incremental effect is smaller by about 0.60 hours. In other words, the search premium associated with high perceived separation risk is concentrated among workers who are not confident they can quickly find a new job after separation.

2.3 From macro pessimism to personal separation risk

How do aggregate expectations map into personal beliefs that are directly relevant for search? Table 3 investigates this link by regressing personal expectation variables on macro unemployment pessimism, controlling for demographics and fixed effects. The estimates reveal a sharp asymmetry: macroeconomic pessimism maps strongly into perceived separation risk but has essentially no association with perceived job-finding probability conditional on job loss.

Workers who are more pessimistic about aggregate unemployment report substantially higher perceived job-loss probability: a worker at the 75th percentile of macro pessimism perceives roughly 4 percentage points higher separation risk than a worker at the 25th percentile. Macro pessimism also correlates with voluntary separation expectations, but the key asymmetry is that it has essentially no relationship with perceived job-finding probability conditional on job loss. This pattern is central for the paper's mechanism: it implies that aggregate pessimism is linked primarily to the perceived *inflow risk* into unemployment rather than perceived duration once unemployed.

Consistent with this pattern, controlling for personal expectations reduces the macro pessimism coefficient in the search regression by about 40 percent. A product-of-coefficients calculation attributes most of this reduction to job-loss expectations, with a secondary role for voluntary-separation beliefs. I use this exercise as descriptive evidence on how aggregate beliefs relate to personal beliefs, not as a causal mediation estimate. In the remainder of the paper, the quantitative model uses the job-loss-risk relationship to evaluate the aggregate implications of precautionary on-the-job search.

2.4 Do beliefs predict realized separations?

A key concern is whether reported job-loss probabilities contain information about future outcomes or instead capture noise or broad sentiment. The SCE's panel structure allows a

Table 3. Macro Expectations and Individual Labor Market Expectations

	(1)	(2)	(3)
	P(Lose Job)	P(Leave Job)	P(Find Job)
$E_t(\Delta \Pr(U_{t+12} \geq 0))$	0.1693*** (0.0062)	0.1362*** (0.0080)	-0.0034 (0.0111)
$E_t(\pi_{t+12})$	0.0001 (0.0003)	-0.0004 (0.0003)	-0.0013*** (0.0004)
Observations	92634	92650	92824
R-squared	0.075	0.060	0.068

Notes: Dependent variables are subjective probabilities (0–1). Regressor is macro unemployment pessimism, $\Delta \Pr(U_{t+12} \geq 0)$, with expected inflation as a control. Sample: employed workers; time and state fixed effects plus demographics; weighted by survey weights; standard errors clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

direct check: do workers who report higher job-loss probability actually separate at higher rates?

Table 4 shows that perceived job-loss risk predicts subsequent separations. A ten percentage point increase in perceived job-loss probability is associated with a 1.3 percentage point higher realized separation rate over the subsequent three months. Although the belief horizon is 12 months, the measure forecasts near-term separations, consistent with persistent differences in separation risk across workers and jobs.⁴ Appendix Table A11 shows that this relationship persists across multiple horizons (1–9 months and censored 12-month outcomes), and Appendix Table A12 shows that it remains positive and significant under both broad non-employment and strict unemployment outcome definitions.

Figure 2, Panel A, provides a nonparametric view. Sorting workers into quintiles of subjective job-loss expectations yields an upward pattern: workers in the top quintile separate at substantially higher rates than workers in the bottom quintile. This evidence is consistent with Stephens (2004) and Mueller et al. (2021), who document that subjective job-loss expectations in household surveys predict actual displacement.

2.5 Mechanism: Evidence of a precautionary motive

The results so far show that perceived job-loss risk predicts search intensity. I now ask whether this relationship reflects precautionary motives—searching to secure job continuity in the event of separation—rather than a generic tendency to search.

⁴The 12-month belief horizon exceeds the 3-month outcome window, but persistent heterogeneity in separation risk across workers implies that beliefs should predict near-term separations even if the stated horizon is longer.

Table 4. Validation: Subjective Job Loss Expectations Predict Separations (Q10-based status)

	(1)	(2)	(3)	(4)	(5)
	None	Controls	Time FE	State + Time FE	State×Time FE
$E_t P(\text{Lose Job}_{t+12})$	0.1547*** (0.0129)	0.1320*** (0.0121)	0.1314*** (0.0120)	0.1307*** (0.0118)	0.1290*** (0.0107)
Controls	No	Yes	Yes	Yes	Yes
Fixed Effects	None	None	Time	State, Time	State×Time
Observations	56412	55993	55993	55832	55832
R-squared	0.021	0.049	0.056	0.059	0.205

Notes: Job-loss expectations are scaled to $[0, 1]$, so coefficients are percentage-point changes in separation probability. Status is constructed from Q10 dummies: employed/attached = FT, PT, leave; unemployed = want work, laid off; NILF = disabled, retiree, student, homemaker. Exact-horizon outcomes require observation at $t + h$ (xtset + F#.). Regressions are weighted by survey weights; SEs clustered by individual.

Stated search motives. The SCE Labor Market Module asks employed workers who report searching to indicate why they are looking for a new job. Table 5 uses this information to test a direct implication of precautionary search: if higher perceived job-loss risk induces search for insurance, it should raise the likelihood that workers cite *job security* as their motive.

Workers with higher perceived job-loss probability are substantially more likely to cite job security as a reason for searching. Voluntary separation expectations, by contrast, enter with the opposite sign: workers who expect to leave voluntarily are *less* likely to cite job security and more likely to cite opportunity-related motives (better pay, advancement). This difference helps separate job-loss expectations from a general desire to move jobs. Finally, conditional on personal expectations, macro pessimism has no independent association with citing job security, suggesting that the security motive operates primarily through beliefs about one's own job-loss risk.

These patterns help distinguish precautionary search from a generic propensity to search: the same beliefs that predict higher search intensity also predict that workers attribute their search to employment stability concerns.

Interaction effects. If workers search to insure against job loss, the payoff to searching while employed should be highest when an outside offer is most useful: when perceived job-loss risk is high and expected re-employment prospects are weak. Table 6 tests this by allowing the effect of macro pessimism on search to vary with (i) perceived job-loss risk

Table 5. Direct Evidence: Why Are Employed Workers Searching?

	(1) Searching for Job Security
$E_t(\text{Pr}(\text{Lose Job}_{t+12}))$	0.588*** (0.071)
$E_t(\text{Pr}(\text{Leave Job}_{t+12}))$	-0.145*** (0.048)
$E_t(\Delta \text{Pr}(U_{t+12} \geq 0))$	-0.026 (0.061)
Time FE	Yes
State FE	Yes
Observations	707
R-squared	0.290

Notes: Outcome equals 1 if the worker reports searching due to job stability concerns or having received a layoff notice. Expectations are scaled to $[0, 1]$, so coefficients are changes in the probability of citing security motives. Sample: employed workers in the SCE Labor Market Module who report searching. All specifications include time and state fixed effects and demographic controls; weighted by survey weights. Standard errors clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

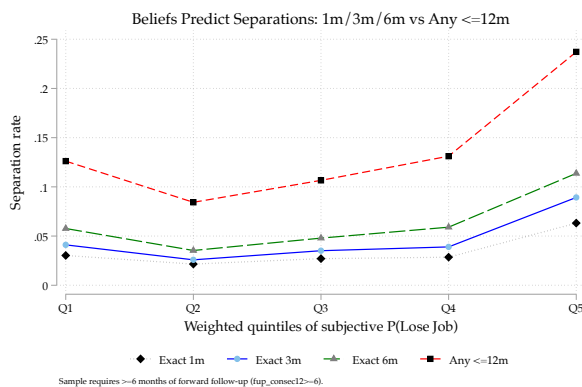
and (ii) perceived re-employment prospects.⁵

Two patterns stand out. First, the interaction between macro pessimism and perceived job-loss probability is positive and statistically significant. Aggregate pessimism induces a larger increase in search among workers who perceive higher personal separation risk.

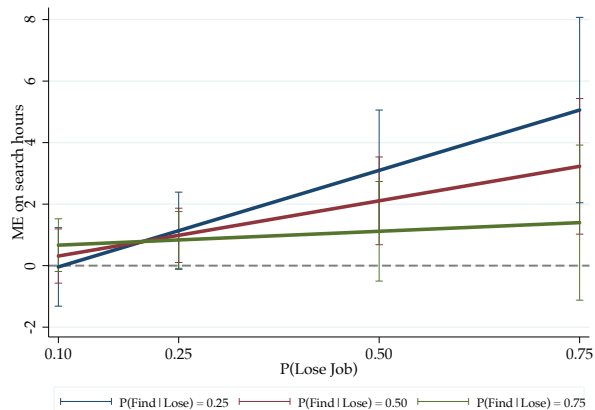
Second, the three-way interaction between macro pessimism, perceived job-loss risk, and perceived job-finding probability is negative and statistically significant. The amplification from job-loss risk is strongest when workers are less confident about finding a job quickly. This is the case in which an outside offer provides the most insurance: high probability of needing it and high expected cost if separation occurs without one.

Formally, the relevant object is the marginal effect of macro pessimism on search, evaluated at different points in the joint distribution of $(\text{Pr}(\text{Lose}), \text{Pr}(\text{Find}))$. Figure 2, Panel B, shows that this marginal effect is largest in the high-risk, low-prospect region, where workers feel vulnerable to job loss and pessimistic about rapid re-employment, and smallest when either separation risk is low or re-employment prospects are strong.

⁵Because the expectations are centered, the main coefficients describe effects at average beliefs, and the interaction terms capture how the response varies across workers.



Panel A. Realized separations by perceived job-loss risk



Panel B. Marginal effect of macro pessimism on search

Notes: Panel A sorts employed workers into quintiles of perceived 12-month job-loss probability and plots realized separation rates at several horizons. Panel B reports marginal effects from the interaction specification in Table 6, using centered expectation variables. It shows how the effect of macro unemployment expectations on search hours varies with perceived job-loss risk and perceived job-finding probability. All models include state×time fixed effects and full controls.

Figure 2. Mechanism Evidence: Beliefs Predict Risk and Shape Search Responses

2.6 Robustness

The baseline estimates document a cross-sectional association between expectations and search. A natural concern is omitted variables: workers in declining industries or weak local labor markets may report both high pessimism and high search intensity because both reflect underlying conditions rather than a behavioral response. The robustness exercises address this and related concerns. Full tables appear in Appendix B.

Alternative fixed effects. Replacing separate state and time fixed effects with state×time interactions absorbs all local aggregate variation, so identification relies on differences across workers within the same state-month cell. Under this specification, the macro pessimism coefficient declines modestly but remains statistically significant, and the job-loss expectations coefficient is essentially unchanged (Table A2). With industry×time fixed effects, which absorb sector-level cyclical conditions, the results are similar (Table A3). These specifications reduce the scope for local or sectoral shocks to explain the estimates.

Functional form and margins. Using alternative transformations of search hours ($\log(1 + y)$, inverse hyperbolic sine) and Poisson specifications leaves the qualitative results intact (Table A5). At the extensive margin (Table A6), personal separation expectations predict whether a worker searches at all, whereas macro pessimism does not once personal expectations are included. Macro beliefs primarily shift intensity among those already searching, while personal job-loss beliefs operate along both margins.

Table 6. Macro Pessimism and Expectations about Personal Outcomes

	(1)	(2)	(3)	(4)	(5)
	Baseline	+ Macro×Lose	+ Macro×Find	+ Lose×Find	Three-way
$E_t(\Delta \Pr(U_{t+12} \geq 0))$	0.906** (0.412)	0.659 (0.405)	0.493 (0.453)	0.452 (0.451)	0.527 (0.414)
$E_t(\Pr(\text{Lose Job}_{t+12}))$	1.841*** (0.502)	1.616*** (0.498)	1.645*** (0.496)	1.694*** (0.499)	1.592*** (0.500)
$E_t(\Pr(\text{Find Job}_{t+3}))$	-1.259*** (0.352)	-1.235*** (0.352)	-1.220*** (0.352)	-1.150*** (0.358)	-1.057*** (0.360)
$E_t(\Delta \Pr(U_{t+12} \geq 0)) \times$ $E_t(\Pr(\text{Lose Job}_{t+12}))$		3.892** (1.794)	3.552* (1.917)	3.445* (1.917)	3.817** (1.772)
$E_t(\Delta \Pr(U_{t+12} \geq 0)) \times$ $E_t(\Pr(\text{Find Job}_{t+3}))$			-0.224 (1.335)	0.036 (1.340)	0.909 (1.259)
$E_t(\Pr(\text{Lose Job}_{t+12})) \times$ $E_t(\Pr(\text{Find Job}_{t+3}))$				-1.364 (1.456)	-0.794 (1.442)
$E_t(\Delta \Pr(U_{t+12} \geq 0)) \times$ $E_t(\Pr(\text{Lose Job}_{t+12})) \times$ $E_t(\Pr(\text{Find Job}_{t+3}))$					-13.438** (5.442)
$E_t(\pi_{t+12})$	-0.023 (0.023)	-0.024 (0.023)	-0.023 (0.023)	-0.024 (0.023)	-0.023 (0.023)
F-stat		4.71			6.10
p-value		0.030			0.014
Observations	4130	4130	4130	4130	4130
Adj. R-squared	0.177	0.179	0.179	0.179	0.181

Notes: Standard errors in parentheses. Dependent variable is weekly job search hours (win-sorized). All specifications include state×time fixed effects and full controls. Standard errors are clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Timing and reverse causality. To address concerns that search may influence expectations rather than the reverse, Table A7 uses lagged expectations from the prior core survey wave, which precedes the LMM by at least one month. The lagged job-loss expectations coefficient remains positive and, if anything, stronger than the contemporaneous estimate. This timing is consistent with expectations preceding search.

Within-person variation. First-difference and individual fixed effects specifications exploit within-worker variation over time (Table A8). The job-loss coefficient remains positive but becomes imprecise, reflecting the short panel dimension: workers appear in the LMM only about 1.4 times on average. Voluntary-separation beliefs remain predictive in some within-person specifications, but this margin is ancillary to the precautionary interpretation emphasized here.

Employed versus unemployed workers. Table A9 compares the role of expectations by employment status. Among employed workers, macro pessimism predicts search intensity. Among the unemployed, the relationship is essentially zero. This asymmetry supports the

paper’s focus on the employed margin: expectations matter for workers who have a job to insure, not for those already unemployed and searching by necessity.

Discussion on identification and scope The estimates in this section document associations, not causal effects. Identifying the causal effect of expectations on search would require exogenous variation in beliefs—variation that shifts what workers think about the labor market without directly affecting their search incentives through other channels. Such variation is difficult to find. Aggregate shocks that change beliefs also change labor market conditions; individual shocks that affect beliefs may also affect search through other mechanisms.

The identifying variation in the baseline analysis is cross-sectional: within the same time and location, workers with more pessimistic expectations search more. A natural question is what generates disagreement in macroeconomic expectations within the same state and month. The predictive content of beliefs, the stated search motives, and the interaction patterns all point in the same direction. But the cross-sectional design cannot rule out unobserved heterogeneity—worker or job characteristics that drive both pessimism and search propensity.

The 2016 election as an illustration. The 2016 presidential election provides a useful illustration of what generates within-cell belief variation. Following [Coibion et al. \(2020\)](#) and [Mian et al. \(2018\)](#), I compare workers in Republican-won and Democrat-won states (dropping swing states) around the November 2016 election. Figure [A3](#) shows event study estimates. The left panel confirms a sharp, persistent divergence in macro unemployment expectations: workers in Republican-won states became approximately 6 percentage points more optimistic relative to those in Democrat-won states. Pre-trends are flat, lending credibility to the parallel trends assumption. The right panel shows the corresponding event study for search hours: relative to Republican-won states, search is lower post-election in Democrat-won states, though the quarterly LMM frequency limits precision.

The election evidence is informative in two ways. First, it identifies one important source of within-cell variation in macro pessimism: partisan affiliation interacted with election outcomes. Workers in the same local labor market can hold sharply different macroeconomic views because of their political priors, and these views shift in response to salient political events. Second, the election shifted macro expectations but did *not* shift personal perceived job-loss probability (Table [A13](#), Column 2). Workers appear to distinguish between their aggregate economic views and their assessment of their own job security—a pattern consistent with the asymmetry documented in Section [2.3](#). Within the election sample, personal job-loss expectations continue to predict search, consistent with

the main analysis. The reduced-form search estimate is imprecise, but its sign implies lower post-election search in Democrat-won states relative to Republican-won states. I treat this evidence as an illustration of the sources of belief disagreement rather than as a stand-alone causal identification strategy.

The results show that expectations about the aggregate economy are related to individual search behavior, and that this link runs mainly through perceived job-loss risk. The aggregate expectations coefficient remains significant even after controlling for personal beliefs, suggesting that macroeconomic pessimism captures perceived risk beyond what the SCE's personal expectation questions span. The model in the next section formalizes this expectations-search link and traces its aggregate implications.

3 A Search-and-Matching Model with Precautionary On-the-Job Search

The empirical analysis documented three facts about employed workers in the Survey of Consumer Expectations. Workers who report higher job-loss risk search more while employed. This response is stronger when expected re-employment prospects are weak and is reflected in job-security search motives. Reported job-loss probabilities also predict subsequent separations.

This section develops a Diamond–Mortensen–Pissarides search-and-matching model with on-the-job search. The model has two main ingredients: separation risk rises in recessions, and employed workers choose search effort before a possible job loss. When a separation occurs, a worker with an outside offer moves directly into a new job next month, while a worker without an offer enters unemployment. Search therefore affects unemployment dynamics by changing the fraction of separations that become unemployment spells.

The model is deliberately parsimonious. It abstracts from voluntary job-to-job transitions absent separation, heterogeneous match quality, and endogenous separations. These simplifications keep the focus on the response of employed search to perceived job-loss risk. Section 4 describes the calibration, and Section 5 uses the calibrated model to quantify the aggregate implications of precautionary on-the-job search.

3.1 Environment and timing

Time is discrete and indexed by $t = 0, 1, 2, \dots$, with each period interpreted as one month. The economy is populated by a unit mass of risk-neutral workers and a continuum of risk-neutral firms. Workers discount future payoffs at factor $\beta \in (0, 1)$. Firms post vacancies at flow cost $\kappa > 0$ and enter freely.

Aggregate states and separations. The aggregate state $z_t \in \{G, B\}$ follows a two-state Markov chain with transition matrix Π , where $\Pi(z, z') \equiv \Pr(z_{t+1} = z' \mid z_t = z)$. Output per filled job is $y(z)$ with $y(G) > y(B)$. Matches are destroyed exogenously at the end of the month with state-dependent hazard $\sigma(z)$, with $\sigma(B) > \sigma(G)$ so that job-loss risk is countercyclical.⁶ With Poisson separations within the month, the corresponding monthly separation probability is

$$\delta(z) \equiv 1 - \exp(-\sigma(z)), \quad 1 - \delta(z) = \exp(-\sigma(z)).$$

This mapping from hazards to probabilities ensures probabilities remain in $[0, 1]$ even when hazards are large.

Timing. Within a period t , the realized aggregate state is z_t . Firms post vacancies and matching occurs. Employed workers choose search intensity $s(z_t)$ and may receive outside offers. Separations are realized at the end of the month. Matches formed during month t become active at $t + 1$ and are therefore not exposed to an additional separation draw within their month of formation. The economy then transitions to z_{t+1} according to Π , and continuation values are evaluated in z_{t+1} .⁷ This timing is convenient empirically: it matches the monthly frequency of the SCE and the notion that current labor-market conditions govern offers and separations over the month, while expectations about the future state matter through discounted continuation values.

Matching technology Let u denote unemployment and v vacancies at the beginning of the month, and define market tightness $\theta \equiv v/u$. Matches are formed according to

$$M(u, v) = \chi u^\alpha v^{1-\alpha}, \quad \alpha \in (0, 1), \chi > 0.$$

⁶The cyclicity of separation risk is well documented; see, e.g., [Fujita and Ramey \(2009\)](#).

⁷Thus, within-month transition probabilities (job finding, offer arrival, separations) are functions of the *current* state z_t , while Π governs only the distribution of next month's continuation values.

The implied contact hazards are

$$\lambda_u(\theta) = \frac{M(u, v)}{u} = \chi \theta^{1-\alpha}, \quad q(\theta) = \frac{M(u, v)}{v} = \chi \theta^{-\alpha}.$$

Tightness summarizes congestion: when θ rises, unemployed workers contact firms more quickly but firms fill vacancies more slowly. Under Poisson arrivals within the month, the job-finding probability for an unemployed worker in state z is

$$p_u(z) = 1 - \exp(-\lambda_u(\theta(z))), \quad (2)$$

and the probability a vacancy is filled in state z is

$$p_f(z) = 1 - \exp(-q(\theta(z))). \quad (3)$$

On-the-job search, offers, and the insurance channel

Search technology. An employed worker chooses search intensity $s \geq 0$ at flow cost

$$c(s) = \frac{\gamma}{2} s^2, \quad \gamma > 0.$$

Convex costs imply an interior optimum whenever the marginal benefit is positive. The parameter γ governs how elastically workers adjust search in response to perceived risk; in the calibration, it is disciplined by the empirical semi-elasticity of search with respect to perceived job-loss risk.

On-the-job search is less efficient than unemployed search. The employed contact hazard is

$$\lambda_e(\theta(z)) = \eta \lambda_u(\theta(z)), \quad \eta \in (0, 1),$$

capturing that employed workers have less time and weaker search incentives per hour than unemployed workers.

Offer arrival. An employed worker who chooses effort s in state z receives outside offers with Poisson hazard

$$\xi(z, s) = \lambda_e(\theta(z)) \psi g(s), \quad (4)$$

where $\psi > 0$ scales search effectiveness and $g(s) = \log(1 + s)$ is increasing and concave. The offer hazard has three components: (i) labor-market conditions through $\lambda_e(\theta(z))$, (ii) a scale parameter ψ governing overall effectiveness of employed search, and (iii)

diminishing returns through $g(s)$. The probability of having an offer available by the end of the month is

$$p_{\text{offer}}(z, s) = 1 - \exp(-\xi(z, s)). \quad (5)$$

Insurance at separation. The effect of search on unemployment is captured by the employment-to-unemployment probability. If separation occurs and no offer is available, the worker enters unemployment next month.⁸ If a separation occurs and the worker holds an offer, she transitions directly to a new job and avoids unemployment. Thus the *objective* employment-to-unemployment probability in state z is

$$\delta_{EU}(z, s) = \delta(z) (1 - p_{\text{offer}}(z, s)). \quad (6)$$

Equation (6) is the key mapping from micro search choices into unemployment inflows: search does not prevent separations, but it reduces the fraction of separations that generate unemployment spells.

Equivalently, the probability of remaining employed next month is

$$\delta_{EE}(z, s) = 1 - \delta_{EU}(z, s) = (1 - \delta(z)) + \delta(z) p_{\text{offer}}(z, s),$$

which makes clear that “staying employed” includes both (i) no separation and (ii) separation covered by an insurance offer.

A broader offer-insurance extension. The mapping in (6) gives outside offers a narrow role: they protect a worker only at the exact moment of a separation. This timing is conservative. In practice, some workers may receive signals or private information about potential job loss before the separation occurs. This is suggested both by the paper’s finding that reported job-loss probabilities predict subsequent separations and by the advance-information logic in [Hendren \(2017\)](#). A worker with such information can search before separation occurs, so an offer can help avoid unemployment even if it is obtained before the separation month.

I capture this with two transparent parameters. Let $\pi \in [0, 1]$ denote the share of eventual separations for which workers receive advance information, and let $K \geq 1$ denote the number of months between that information and the separation. Workers without

⁸This assumption separates the insurance role of offers from the job-ladder motive in models such as [Postel-Vinay and Robin \(2002\)](#), where workers accept outside offers to climb toward better-paying jobs. Abstracting from voluntary job-to-job transitions keeps the focus on the precautionary motive suggested by the SCE evidence on search motives.

advance information have the baseline one-month coverage probability $p_{\text{offer}}(z, s)$. Workers with advance information can search for K months before the separation occurs, so their cumulative probability of obtaining at least one offer is $1 - (1 - p_{\text{offer}}(z, s))^K$. The protected share of separations is

$$a(z, s; \pi, K) = (1 - \pi)p_{\text{offer}}(z, s) + \pi \left[1 - (1 - p_{\text{offer}}(z, s))^K \right], \quad (7)$$

and the employment-to-unemployment probability becomes

$$\delta_{EU}^{\pi, K}(z, s) = \delta(z) [1 - a(z, s; \pi, K)]. \quad (8)$$

The baseline corresponds to $\pi = 0$ or $K = 1$. For intuition, define the effective protection factor

$$\omega_{\text{eff}}(z, s; \pi, K) = \frac{a(z, s; \pi, K)}{p_{\text{offer}}(z, s)}.$$

When offer probabilities are small, $\omega_{\text{eff}} \approx 1 + \pi(K - 1)$. In the calibrated model offer probabilities are not tiny, so I use the exact expression in (7).

Discipline and identification. I choose π and K using the SCE job supplement. Among E-E transitions due to job loss or anticipated layoff, 47.5 percent report advance notice; among those with notice, the average notice period is 2.77 months. These moments determine how much of an offer arrival can protect a future separation before it becomes an unemployment spell, while the empirical estimates above continue to discipline the search response.

Effect on optimal search. The search first-order condition retains its form with $a(z, s; \pi, K)$ replacing $p_{\text{offer}}(z, s)$ inside the marginal-benefit term. The relevant marginal term is

$$\frac{\partial a(z, s; \pi, K)}{\partial p_{\text{offer}}} = (1 - \pi) + \pi K (1 - p_{\text{offer}}(z, s))^{K-1}.$$

Thus advance information raises the value of search by giving some workers more time to obtain an offer before separation occurs. The sequence is the same as in the baseline: macro pessimism is related to perceived job-loss risk, perceived risk raises search, and search generates offers that reduce unemployment inflows.

Section 5 reports the quantitative implications. The narrow timing of the baseline is a conservative choice: precautionary on-the-job search contributes more to unemployment inflows when workers receive advance information and can search before a separation is realized.

The model abstracts from voluntary job-to-job transitions when no separation occurs;

offers are exercised only upon separation to isolate the insurance margin.⁹ This assumption rules out wage-ladder reallocation and the procyclicality of employer-to-employer flows as objects the model can match. The benefit is a transparent mapping from search to unemployment inflows: search is chosen to insure against *involuntary* transitions into unemployment.

Beliefs Workers form subjective beliefs about job-loss risk. Let perceived separation hazard be

$$\tilde{\sigma}(z) = \zeta \cdot m(z) \cdot \sigma(z), \quad (9)$$

where $\zeta > 0$ is a level shifter and $m(z)$ allows for state-dependent belief distortions. In the baseline calibration, $m(G) = m(B) = 1$, so perceived risk inherits its cyclicity from the objective hazard. Workers behave as if the separation probability were

$$\tilde{\delta}(z) \equiv 1 - \exp(-\tilde{\sigma}(z)).$$

Beliefs affect behavior through search; realized separations are governed by $\sigma(z)$. This distinction aligns with the empirical design: subjective expectations predict search choices even though objective separations are realized according to data-consistent hazards.

3.2 Value functions, wages, and equilibrium

Let $W(z)$ and $U(z)$ denote the worker's values of employment and unemployment in state z , computed under the worker's subjective beliefs, and let $J(z)$ denote the firm's value of a filled job under the objective separation process.

Unemployed worker. An unemployed worker receives flow payoff b and finds a job within the month with probability $p_u(z)$, determined by current tightness. Continuation values are evaluated next month after the aggregate transition. The value of unemployment satisfies

$$U(z) = b + \beta \sum_{z'} \Pi(z, z') \left[p_u(z) W(z') + (1 - p_u(z)) U(z') \right], \quad (10)$$

Employed worker. An employed worker receives wage $w(z)$, chooses search intensity $s(z)$, and faces separation risk within the month. The *perceived* employment-to-

⁹In job-ladder models, employed workers accept offers without separation; see [Postel-Vinay and Robin \(2002\)](#). The present model shuts down that margin to focus on insurance against unemployment inflows.

unemployment probability is

$$\tilde{\delta}_{EU}(z, s) \equiv \tilde{\delta}(z) (1 - p_{\text{offer}}(z, s)).$$

Then the value of employment satisfies

$$W(z) = w(z) - c(s(z)) + \beta \sum_{z'} \Pi(z, z') \left[(1 - \tilde{\delta}_{EU}(z, s(z))) W(z') + \tilde{\delta}_{EU}(z, s(z)) U(z') \right]. \quad (11)$$

This equation highlights the precautionary tradeoff: the worker pays a current cost $c(s)$ to reduce the perceived probability of entering unemployment next month, thereby preserving the expected employment surplus $W - U$. This is where the precautionary motive enters: higher perceived separation risk raises the probability of needing an offer, increasing the marginal value of search effort.

Firm. A filled job produces $y(z)$ this month and pays wage $w(z)$. The match survives the month with probability $\exp(-\sigma(z))$ and then the aggregate state transitions to z' . The value of a filled job is

$$J(z) = y(z) - w(z) + \beta \exp(-\sigma(z)) \sum_{z'} \Pi(z, z') J(z'). \quad (12)$$

Because offers are exercised only upon exogenous separation, employed search does not create an additional quit risk for firms; the only destruction margin is $\sigma(z)$.¹⁰ This keeps the firm problem standard and ensures the worker's search decision affects unemployment primarily through flows on the worker side.

Wages. Wages follow a benchmark DMP Nash wage with free entry:

$$w(z) = (1 - \phi)b + \phi(y(z) + \kappa\theta(z)), \quad (13)$$

where ϕ is the worker's bargaining weight. The wage is increasing in productivity and in market tightness (which raises the worker's outside option through improved job-finding prospects). The wage does not respond to the worker's on-the-job search choice, keeping the focus on the insurance channel: employed search affects unemployment risk

¹⁰Firms do not internalize the worker's search effort: because offers are exercised only upon exogenous separation, employed search does not affect the firm's continuation value. If workers could voluntarily accept outside offers (the job-ladder margin), the firm would face an additional source of match destruction. Abstracting from this channel keeps the firm problem standard and isolates the insurance mechanism.

through the separation-to-unemployment mapping, not through within-job wage renegotiation. The interpretation is contract-like wage setting (or infrequent renegotiation): employed search affects labor-market dynamics through flows and insurance, not through contemporaneous wage renegotiation.

Vacancy creation. Free entry implies that the expected value of posting a vacancy is zero. A vacancy posted in state z is filled within the month with probability $p_f(z)$, and then yields continuation value $J(z')$ next month after the aggregate transition. The free-entry condition is

$$\kappa = \beta p_f(z) \sum_{z'} \Pi(z, z') J(z'), \quad (14)$$

This condition determines tightness $\theta(z)$ jointly with job values: in booms, higher output and lower separation risk raise J , inducing vacancy entry and higher θ .

Optimal on-the-job search. The employed worker chooses $s(z)$ to maximize $W(z)$. Differentiating (11) with respect to s yields

$$c'(s(z)) = \beta \tilde{\delta}(z) \frac{\partial p_{\text{offer}}(z, s(z))}{\partial s} \sum_{z'} \Pi(z, z') (W(z') - U(z')), \quad (15)$$

where

$$\frac{\partial p_{\text{offer}}(z, s)}{\partial s} = \exp(-\xi(z, s)) \cdot \lambda_e(\theta(z)) \psi g'(s), \quad g'(s) = \frac{1}{1+s}.$$

The right-hand side is the marginal insurance value of search. It is high when (i) perceived separation risk is high (the worker is more likely to need an insurance offer), (ii) labor markets are tight (search is more effective at generating offers), and (iii) the employment surplus is large (avoiding unemployment is valuable). This decomposition will be useful in the counterfactuals: search can be countercyclical if the rise in perceived risk in recessions dominates the fall in search effectiveness from lower tightness.

Equilibrium. A Markov equilibrium is a collection $\{W(z), U(z), J(z), \theta(z), s(z), w(z)\}_{z \in \{G, B\}}$ satisfying (10)–(15) and (12)–(14) in each state.

3.3 Unemployment dynamics

Given the equilibrium policy $s(\cdot)$, unemployment evolves according to objective flow probabilities. Under the timing in Section 3.1, within-month job finding, offer arrival,

and separations are realized in the *current* state z_t , while the Markov transition $\Pi(z_t, z_{t+1})$ governs only next month's continuation values. The law of motion for unemployment is therefore

$$u_{t+1} = u_t(1 - p_u(z_t)) + (1 - u_t) \delta_{EU}(z_t, s(z_t)),$$

where $\delta_{EU}(z, s)$ is given by (6). If the economy were held fixed in state z , the flow-balance condition implies a steady-state unemployment rate of

$$u^{ss}(z) = \frac{\delta_{EU}(z, s(z))}{\delta_{EU}(z, s(z)) + p_u(z)}. \quad (16)$$

Precautionary search affects unemployment primarily through the inflow margin: when perceived job-loss risk rises, workers search more, which raises $p_{\text{offer}}(z, s)$ and reduces the fraction of separations that translate into unemployment. The job-finding probability $p_u(z)$ responds indirectly in general equilibrium through market tightness, but employed search does not enter $p_u(z)$ directly.

3.4 How perceived risk drives search and stabilizes unemployment

The next results formalize the mechanism and motivate the counterfactual exercises. Proofs are in Appendix C.7. For reference, the offer probability and its derivative in current state z are

$$p_{\text{offer}}(z, s) \equiv 1 - \exp\left(-\lambda_e(\theta(z)) \psi g(s)\right), \quad (17)$$

$$p_{\text{offer},s}(z, s) \equiv \frac{\partial p_{\text{offer}}(z, s)}{\partial s} = \exp\left(-\lambda_e(\theta(z)) \psi g(s)\right) \cdot \lambda_e(\theta(z)) \psi g'(s) > 0, \quad (18)$$

and the objective employment-to-unemployment probability is

$$\delta_{EU}(z, s) \equiv \delta(z) (1 - p_{\text{offer}}(z, s)), \quad (19)$$

with $\tilde{\delta}(z) = 1 - \exp(-\tilde{\sigma}(z))$ and $\tilde{\sigma}(z)$ defined in (9).

Assumption 1 (Regularity). (i) $c(\cdot)$ is twice continuously differentiable, strictly convex, and satisfies $c'(0) = 0$. (ii) $g(\cdot)$ is twice continuously differentiable, increasing, and concave. (iii) The expected employment surplus is positive: $\sum_{z'} \Pi(z, z') (W(z') - U(z')) > 0$ for each z .

Proposition 1 (Precautionary on-the-job search). Under Assumption 1, holding fixed $\theta(\cdot)$, Π , and the continuation surplus $\sum_{z'} \Pi(z, z') (W(z') - U(z'))$, the optimal search intensity $s(z)$ is

increasing in perceived job-loss risk in that state:

$$\frac{\partial s(z)}{\partial \tilde{\delta}(z)} > 0 \quad \text{for } z \in \{G, B\}.$$

Proposition 1 is the model's core result: workers search harder when they perceive higher separation risk, formalizing the empirical relationship documented in Section 2. The intuition follows directly from the FOC: higher $\tilde{\delta}(z)$ raises the marginal benefit of search by increasing the probability that the worker will need an outside offer.

Proposition 2 (Search provides insurance). *Under Assumption 1, the objective employment-to-unemployment probability $\delta_{EU}(z, s)$ is strictly decreasing in s and strictly increasing in $\sigma(z)$:*

$$\frac{\partial \delta_{EU}(z, s)}{\partial s} < 0, \quad \frac{\partial \delta_{EU}(z, s)}{\partial \sigma(z)} > 0.$$

Proposition 2 confirms that more search reduces the rate at which separations translate into unemployment.

Proposition 3 (Conditions for countercyclical search). *Under Assumption 1, define the perceived marginal benefit of search in state z given effort s as*

$$\mathcal{MB}(z; s) \equiv \tilde{\delta}(z) p_{\text{offer},s}(z, s) \sum_{z'} \Pi(z, z') (W(z') - U(z')).$$

If $\mathcal{MB}(B; s) \geq \mathcal{MB}(G; s)$ for all s in a neighborhood containing $\{s(G), s(B)\}$, then $s(B) \geq s(G)$.

Whether on-the-job search is countercyclical depends on which force dominates: recessions raise the precautionary motive through $\tilde{\delta}(B)$ but typically reduce the return to effort through lower tightness (entering $p_{\text{offer},s}$) and potentially lower surplus.

Corollary 1 (Constant- σ counterfactual). *Consider a counterfactual economy with constant separation hazard $\sigma(G) = \sigma(B) \equiv \bar{\sigma}$ and no belief distortions $m(G) = m(B) = 1$. Then $\tilde{\delta}(G) = \tilde{\delta}(B)$, so cyclical variation in the precautionary motive is removed. Any remaining cyclicity of $s(\cdot)$ is driven by differences in offer arrival (through $\theta(\cdot)$) and differences in the employment surplus.*

Proposition 4 (Precautionary search dampens unemployment in recessions). *Consider a recession state B . Let $s^{\text{base}}(B)$ denote the equilibrium search intensity in state B and let $s^{\text{fix}}(B) \equiv s(G)$ denote search held fixed at its expansion level. If $s^{\text{base}}(B) > s^{\text{fix}}(B)$, then unemployment inflows are lower in the baseline:*

$$\delta_{EU}(B, s^{\text{base}}(B)) < \delta_{EU}(B, s^{\text{fix}}(B)).$$

Consequently, holding fixed $p_u(B)$, steady-state unemployment in state B is lower under baseline search than under fixed search.

This is the paper’s central aggregate result. When workers endogenously raise search effort in recessions, more of them hold outside offers at the moment of separation, converting what would have been unemployment spells into job-to-job transitions. The resulting reduction in unemployment inflows dampens the recessionary rise in unemployment.

Remark 1 (Baseline mechanism does not rely on belief distortions). In the baseline calibration, $m(G) = m(B) = 1$. Therefore, perceived job-loss risk inherits its cyclicalities from the objective separation hazard. Countercyclical separation risk can generate higher search effort in bad times even without state-dependent belief distortions.

3.5 Limitations and extensions

Several modeling choices deserve comment.

Why abstract from job-to-job transitions? The model assumes that offers are exercised only upon separation, so there are no voluntary job-to-job transitions. In models with a job ladder (Postel-Vinay and Robin, 2002; Moscarini and Postel-Vinay, 2018), workers search to climb toward better-paying jobs, and the cyclicalities of job-to-job flows reflect variation in match quality and outside options. By shutting down this margin, the present model isolates the precautionary channel. The cost is that the model cannot speak to the procyclicality of employer-to-employer flows or the role of match quality in search decisions.

Why exogenous separations? Separation is governed by the exogenous hazard $\sigma(z)$ rather than an endogenous destruction margin. This keeps the model tractable and focuses attention on the worker’s search response to perceived risk, which is the object disciplined by the data. A richer model with endogenous separations (Fujita and Ramey, 2009; Pries, 2008) could generate additional feedback between search and match dissolution, but at the cost of introducing margins that are difficult to identify separately in the SCE data.

The next section calibrates the model to match moments from the Survey of Consumer Expectations and the broader U.S. labor market.

3.6 Extension: Separating job-ladder and precautionary motives

The baseline model isolates the precautionary motive for on-the-job search by assuming that outside offers are exercised only when an exogenous separation occurs. This assumption is useful for transparency, but it shuts down the standard job-ladder motive

emphasized in models of employer-to-employer mobility. This subsection shows how the model can be extended to allow outside offers to serve two distinct roles: a *job-ladder role* when the worker is not separated, and a *precautionary insurance role* when the worker is separated.

The key distinction is timing. An employed worker searches during the month and may receive an outside offer with probability $p_{\text{offer}}(z, s)$. If the worker is not separated, the offer may be used to move voluntarily to a better job. If the worker is separated, the offer may be used to avoid unemployment. Thus the same search effort generates two conceptually different payoffs.

Let $\rho_J \in [0, 1]$ denote the probability that an outside offer is acceptable as a voluntary job-ladder move when no separation occurs. Let $\rho_I \in [0, 1]$ denote the probability that an outside offer is usable as insurance when a separation occurs. The baseline model corresponds to the special case $\rho_J = 0$ and $\rho_I = 1$.

To capture the value of a voluntary job-to-job move in a parsimonious way, suppose that an acceptable ladder offer yields a continuation gain $\Delta_J(z') \geq 0$ relative to remaining in the current job next period. This reduced-form gain summarizes better pay, better match quality, amenities, or career advancement. The formulation avoids introducing a full wage-offer or match-quality distribution while still allowing search to have the standard job-ladder payoff.

With these objects, the employed worker's value becomes

$$\begin{aligned}
 W(z) = w(z) - c(s(z)) + \beta \sum_{z'} \Pi(z, z') & \left\{ (1 - \tilde{\delta}(z)) \left[\rho_J p_{\text{offer}}(z, s(z)) (W(z') + \Delta_J(z')) \right. \right. \\
 & \left. \left. + (1 - \rho_J p_{\text{offer}}(z, s(z))) W(z') \right] \right. \\
 & \left. + \tilde{\delta}(z) \left[\rho_I p_{\text{offer}}(z, s(z)) W(z') + (1 - \rho_I p_{\text{offer}}(z, s(z))) U(z') \right] \right\}.
 \end{aligned} \tag{20}$$

The first bracket is the no-separation branch. With perceived probability $1 - \tilde{\delta}(z)$, the worker keeps the current job unless she receives and accepts a ladder offer. Conditional on an acceptable ladder offer, she obtains $W(z') + \Delta_J(z')$ rather than $W(z')$. The second bracket is the separation branch. With perceived probability $\tilde{\delta}(z)$, the worker loses the current job. If an offer is usable as insurance, she remains employed next period; otherwise she enters unemployment.

Differentiating (20) with respect to s gives the search first-order condition:

$$c'(s(z)) = \beta p_{\text{offer},s}(z, s(z)) \sum_{z'} \Pi(z, z') \left[\underbrace{(1 - \tilde{\delta}(z)) \rho_J \Delta_J(z')}_{\text{job-ladder motive}} + \underbrace{\tilde{\delta}(z) \rho_I (W(z') - U(z'))}_{\text{precautionary insurance motive}} \right]. \quad (21)$$

Equation (21) separates the two motives for on-the-job search. The first term is the job-ladder motive: search is valuable because it may generate a voluntary move to a better job even if the current match survives. This motive is weighted by the probability of no separation, $1 - \tilde{\delta}(z)$, and by the acceptance probability ρ_J . The second term is the precautionary motive: search is valuable because an offer can prevent unemployment if the current match is destroyed. This motive is weighted by perceived separation risk, $\tilde{\delta}(z)$, and by the usability of an offer as insurance, ρ_I .

This decomposition clarifies the paper's mechanism. Standard job-ladder search is valuable primarily when outside offers improve the worker's continuation value. Precautionary search is valuable when separation risk is high and unemployment is costly. The two motives may have different cyclical properties. The job-ladder component is likely to be stronger in expansions, when tight labor markets make outside offers more plentiful and valuable. The precautionary component may be stronger in recessions, when perceived separation risk rises. Countercyclical employed search arises when the increase in the precautionary component dominates the decline in offer-arrival effectiveness and any weakening of the job-ladder component.

The objective worker-flow equations also separate cleanly. In the extended model, the employment-to-unemployment probability is

$$\delta_{EU}(z, s) = \delta(z) (1 - \rho_I p_{\text{offer}}(z, s)). \quad (22)$$

Search reduces unemployment inflows only through the insurance component. Voluntary ladder offers do not directly affect the employment-to-unemployment margin.

The model now generates two types of employment-to-employment transitions. The first are voluntary job-ladder moves among workers whose current match survives:

$$\delta_{EE}^J(z, s) = (1 - \delta(z)) \rho_J p_{\text{offer}}(z, s). \quad (23)$$

The second are insured separations: workers whose current match is destroyed but who

avoid unemployment because they have a usable outside offer:

$$\delta_{EE}^I(z, s) = \delta(z) \rho_I p_{\text{offer}}(z, s). \quad (24)$$

Total employment-to-employment flows in the model are therefore

$$\delta_{EE}^{\text{total}}(z, s) = \delta_{EE}^J(z, s) + \delta_{EE}^I(z, s). \quad (25)$$

The distinction between δ_{EE}^J and δ_{EE}^I is important empirically. The first object corresponds to ordinary job-ladder mobility. The second object corresponds to covered separations that would have become employment-to-unemployment transitions absent precautionary search. The baseline model sets $\rho_J = 0$, so all employment-to-employment transitions generated by the model are insured separations.

Because voluntary job-ladder moves create quit risk for firms, the firm's value must be adjusted relative to the baseline model. In the extended model, a filled job survives to the next period only if the match is not exogenously destroyed and the worker does not voluntarily leave for an acceptable ladder offer. The firm value becomes

$$J(z) = y(z) - w(z) + \beta \exp(-\sigma(z)) \left(1 - \rho_J p_{\text{offer}}(z, s(z))\right) \sum_{z'} \Pi(z, z') J(z'). \quad (26)$$

The insurance parameter ρ_I does not enter the firm's survival probability because insurance offers are used only after the match has already been destroyed. Free entry remains

$$\kappa = \beta p_f(z) \sum_{z'} \Pi(z, z') J(z'), \quad (27)$$

with $J(z)$ now given by (26). Wages may be kept at the same benchmark Nash wage in (13); doing so preserves the interpretation that the extension adds flow motives for search without introducing wage renegotiation over outside offers.

The extended model nests several useful counterfactuals. Setting $\rho_J = 0$ recovers the insurance-only model used in the baseline. Setting $\rho_I = 0$ shuts down the precautionary use of offers while preserving ordinary job-ladder search. Holding $\tilde{\delta}(z)$ fixed across states removes the cyclical precautionary motive while preserving objective separations in the flow equations. Varying ρ_I between zero and one measures how much the aggregate stabilization depends on the assumption that outside offers are usable at separation. These counterfactuals allow the quantitative exercise to decompose employed search into its job-ladder and precautionary components.

4 Calibration

This section calibrates the model to assess the quantitative relevance of the precautionary search mechanism. The calibration separates two sets of moments. Aggregate labor-market stocks and flows discipline the matching block and the unemployment law of motion. Moments from the Survey of Consumer Expectations discipline the belief and search components of the model. This separation is useful because the model's unemployment dynamics obey a stock-flow accounting identity, while the SCE is most informative about subjective job-loss risk, offer receipt, and the behavioral response of search to perceived risk.

The main calibration therefore uses an aggregate stock-flow benchmark for unemployment and unemployment-to-employment transitions. It uses the SCE to discipline mean perceived job-loss risk, the on-the-job offer probability, and the semi-elasticity of search with respect to perceived job-loss risk. The perceived job-loss target is computed in the employed SCE Labor Market Module sample used in the search regressions, rather than in the broader core-survey employed sample. SCE transition rates are reported as validation moments rather than imposed as hard targets. This avoids mixing transition rates from a rotating survey panel with aggregate unemployment stocks in a way that violates flow balance.

4.1 Externally Set Parameters

Table 7 reports externally set parameters. Time is monthly. Aggregate productivity follows a two-state Markov process $z \in \{G, B\}$ with productivity levels $y(G) = 1.00$ and $y(B) = 0.95$, implying a 5 percent gap between expansions and recessions. The transition matrix is disciplined by NBER business-cycle durations: an average expansion of 58 months and an average recession of 11 months imply monthly persistence probabilities $\Pi_{GG} = 0.99$ and $\Pi_{BB} = 0.90$. The ergodic distribution assigns probability 0.91 to expansions and 0.09 to recessions.

Workers discount the future at $\beta = 0.9967$, corresponding to an annual discount factor of 0.96. Wages are determined by Nash bargaining with worker share $\phi = 0.5$. The matching elasticity with respect to unemployment is $\alpha = 0.5$, consistent with [Petrongolo and Pissarides \(2001\)](#). The flow value of unemployment is $b = 0.4$, implying a replacement rate of roughly 40 percent of the mean wage.

Job separations follow a state-dependent Poisson process. The recession-expansion separation ratio is set to match the cyclicalities of layoffs and discharges in the data, with $\sigma(B)/\sigma(G) \approx 1.7$, consistent with the cyclicalities of separations documented in [Fujita and](#)

Ramey (2009). In the main calibration, the average separation level is disciplined by the employment-to-unemployment flow, while the recession-expansion ratio controls the cyclicity of separation risk. Employed workers face a lower contact rate than unemployed workers, $\lambda_e(z) = \eta \lambda_u(z)$, with $\eta = 0.1$.¹¹

Table 7. Externally Set Parameters

Parameter	Description	Value	Source
<i>Preferences</i>			
β	Monthly discount factor	0.9967	4% annual rate
b	Flow value of unemployment	0.40	UI benefits; Shimer (2005)
<i>Matching</i>			
α	Matching elasticity with respect to unemployment	0.50	Petrongolo and Pissarides (2001)
ϕ	Worker bargaining share	0.50	Hosios condition
η	Employed/unemployed contact ratio	0.10	Faberman et al. (2022)
<i>Separations</i>			
$\sigma(B)/\sigma(G)$	Recession-expansion separation ratio	1.67	Layoffs and discharges; Fujita and Ramey (2009)
<i>Aggregate states</i>			
$y(G), y(B)$	Productivity	1.00, 0.95	5% output gap
Π_{GG}, Π_{BB}	State persistence	0.99, 0.90	NBER durations
<i>Beliefs</i>			
$m(G), m(B)$	Belief multipliers	1.0, 1.0	No state-dependent distortions

Notes: Parameters are set externally from data or prior literature. The separation ratio governs the cyclicity of separation risk. The level of separation risk is disciplined jointly with the aggregate flow targets in the internal calibration.

4.2 Internally Calibrated Parameters

The main calibration chooses six parameters: matching efficiency χ , vacancy posting cost κ , search effectiveness ψ , search cost curvature γ , the belief level shifter ζ , and the

¹¹The parameter η governs market-level contact rates. Faberman et al. (2022) find that employed workers are more effective than unemployed workers in converting search effort into offers. The distinction is that η captures the lower time available for search among employed workers, while the parameter ψ scales the effectiveness of search effort conditional on searching.

expansion separation hazard $\sigma(G)$. The recession separation hazard is then given by $\sigma(B) = r_\sigma \sigma(G)$, where r_σ is the externally set recession-expansion separation ratio.

The calibration uses six primary moments: the aggregate unemployment benchmark, the monthly unemployment-to-employment probability, the stock-flow employment-to-unemployment benchmark implied by those two moments, mean perceived job-loss risk in the SCE, the employed offer probability in the SCE, and the semi-elasticity of search with respect to perceived job-loss risk. The baseline model's effective employment-to-unemployment probability is then reported after outside offers insure some separations. Table 8 reports these moments and calibrated values.

Aggregate stock-flow block. The aggregate unemployment benchmark and the monthly unemployment-to-employment probability discipline the matching and separation blocks. These moments imply a benchmark employment-to-unemployment flow through the steady-state flow-balance condition

$$u = \frac{P_{EU}}{P_{EU} + P_{UE}}.$$

For this reason, the main calibration keeps the aggregate stock-flow benchmark separate from the SCE validation flows. Matching efficiency χ and vacancy cost κ govern job finding and vacancy creation. The separation level $\sigma(G)$, together with the fixed ratio $\sigma(B)/\sigma(G)$, governs the inflow risk faced by employed workers. In the baseline economy with precautionary search, effective employment-to-unemployment flows are lower than the raw separation benchmark because some separations are covered by outside offers.

This structure keeps the unemployment accounting transparent. A higher χ raises contact rates for given tightness. A higher κ lowers vacancy creation and reduces market tightness. A higher separation level raises employment-to-unemployment flows and, through flow balance, raises unemployment for a given job-finding rate.

Beliefs. Workers perceive job-loss risk according to

$$\tilde{\sigma}(z) = \zeta \cdot m(z) \cdot \sigma(z),$$

where ζ scales the level of perceived risk and $m(z)$ allows perceived risk to vary with the aggregate state beyond the movement in actual separations. In the benchmark calibration, I set $m(G) = m(B) = 1$. Thus perceived risk rises in recessions because actual separation risk rises. The parameter ζ is chosen to match the mean subjective probability of job loss over the next 12 months in the SCE.

I also report a recession-belief version that disciplines $m(B) > m(G)$ using the empirical relationship between macro pessimism and workers' perceived job-loss risk. Table 3 implies that a one-standard-deviation increase in macro pessimism raises perceived job-loss risk by about 4.0 percentage points. In that version, $m(G) = 1$, $m(B) = 1.17$, and $\zeta = 0.759$. Actual separation risk still determines realized unemployment inflows; perceived separation risk determines workers' search choices.

Search technology. Employed workers choose search intensity s with flow disutility

$$c(s) = \frac{\gamma}{2}s^2.$$

Offer arrival follows a Poisson process,

$$p_{\text{offer}}(z, s) = 1 - \exp(-\lambda_e(\theta(z)) \psi g(s)), \quad g(s) = \log(1 + s).$$

The parameter ψ scales the effectiveness of employed search in generating offers, while γ governs the cost of search effort.

Two SCE moments discipline this block. The first is the employed offer probability, which pins down the average effectiveness of search. The second is the semi-elasticity of search with respect to perceived job-loss risk,

$$\beta_{\text{plose}} = \frac{\partial \mathbb{E}[\text{SearchHours} \mid \tilde{p}^{12m}]}{\partial \tilde{p}^{12m}}, \quad (28)$$

where perceived job-loss risk is measured on a 0–1 scale. In the model, this object is computed by perturbing perceived job-loss risk around the calibrated value. A larger value of β_{plose} requires lower marginal search costs, so that workers respond more strongly to changes in perceived separation risk.

I do not target the level of search hours. The model uses a reduced-form effort variable, while the data measure time spent searching. Matching the level would require an arbitrary units conversion. Instead, the calibration targets the behavioral object that matters for the mechanism: how strongly employed search responds to perceived job-loss risk.

Appendix Table A14 reports the complete parameterization and implied steady-state objects used in the quantitative exercises.

Discussion of calibrated values. The calibrated belief parameter ζ maps objective separation risk into the average perceived job-loss risk reported in the SCE search sample. With $m(G) = m(B) = 1$, the benchmark does not require an additional state-dependent

Table 8. Calibration Targets and Parameter Values

Panel A. Calibration Targets			
Moment	Data	Model	Role
<i>A. Aggregate stock-flow targets</i>			
Adjusted SCE unemployment rate u	0.057	0.056	target
Monthly U-to-E probability p_{UE}	0.280	0.287	target
Stock-flow E-to-U probability p_{EU}	0.017	0.017	implied by u and p_{UE}
<i>B. SCE behavior and expectations targets</i>			
Mean perceived job-loss probability, 12 months	0.212	0.212	target
Employed offer probability	0.330	0.333	target
Search semi-elasticity β_{plose}	1.688	1.688	target
<i>C. Validation moments</i>			
SCE monthly U-to-E probability	0.200	0.287	validation
SCE monthly E-to-U probability	0.0116	0.0169	validation
SCE monthly total E-to-E probability	0.0385	0.0085	validation
Covered E-to-E share of total E-to-E		0.221	validation
Panel B. Calibrated Parameters			
Parameter	Description	Value	
χ	Matching efficiency	0.338	
κ	Vacancy cost	0.508	
ψ	Search effectiveness	26.282	
γ	Search cost curvature	0.023	
ζ	Belief level shifter	0.775	
$m(G), m(B)$	State belief multipliers	1.000, 1.000	
$\sigma(G), \sigma(B)$	Separation hazards	0.024, 0.040	

Notes: The main calibration separates aggregate stock-flow targets from SCE behavior and expectations targets. The unemployment target uses the SCE labor-force measure that excludes temporary layoffs from unemployment. The stock-flow E-to-U probability is implied by the adjusted unemployment rate and the monthly U-to-E probability. Because outside offers insure some separations, the calibration grosses up the exogenous separation hazard so that the effective E-to-U flow matches this stock-flow target. SCE moments discipline perceived job-loss risk, offer receipt, and the response of search to perceived risk. SCE transition rates are reported as validation moments. The model-implied E-to-E flow is a covered-separation flow, not total employer-to-employer mobility, because the baseline model abstracts from voluntary job-ladder transitions.

belief wedge to generate countercyclical search. The comparison below then asks how much the aggregate implication changes when the estimated macro-to-personal belief movement is added.

The search effectiveness parameter ψ determines the average probability that an employed worker obtains an offer for a given level of effort and market tightness. The search cost parameter γ determines how strongly search effort responds to changes in perceived job-loss risk. Together, these parameters govern the strength of the insurance channel: workers must be able to obtain outside offers, and they must increase search when separation risk rises.

The separation hazards determine the objective risk that employment relationships end. In the baseline model, offers do not prevent separations; they only change what happens after a separation occurs. This distinction is central. The model-implied employment-to-unemployment probability is

$$p_{EU}(z) = \delta(z)(1 - p_{\text{offer}}(z, s(z))),$$

while the covered-separation flow is

$$p_{EE}^I(z) = \delta(z)p_{\text{offer}}(z, s(z)).$$

Thus search lowers unemployment inflows by increasing the share of separations covered by outside offers.

4.3 Untargeted Model Implications

I now evaluate implications that are not targeted in calibration. The targeted moments discipline aggregate unemployment accounting, mean perceived job-loss risk, the employed offer rate, and the semi-elasticity of search with respect to perceived job-loss risk. The moments below are held out.

The objective is to assess whether the data line up with the mechanism emphasized in the empirical section: higher perceived separation risk predicts more on-the-job search, more search predicts more offer receipt, and offer receipt is associated with lower subsequent unemployment risk. These validation moments are not used to choose parameters.

Table 9 reports four untargeted moments from the SCE employed sample using survey weights. The pattern is consistent with the mechanism. Workers with high perceived job-loss risk search more: search hours are 4.76 in the top risk quintile and 3.68 in the bottom quintile. Offer receipt is also higher among workers who search more intensively:

Table 9. Model Validation: Untargeted Moments from SCE

Moment (employed sample)	Low	High	Diff.
Search hours by perceived job-loss quintile	3.68 (Q1)	4.76 (Q5)	+1.08
Offer rate by search-hours quintile	25.0% (Q1)	44.0% (Q5)	+19.0 pp
Offer rate by perceived job-loss quintile	13.2% (Q1)	16.9% (Q5)	+3.6 pp
Next-wave unemployment by offer receipt	0.93% (no offer)	0.85% (offer)	-0.08 pp

Notes: Weighted means from the SCE employed sample. “Next-wave unemployment” uses F1_unemp. For the offer-by-search moment, the available overlap with offer data uses search_hours_raw from the job supplement. No regression adjustment is imposed in this table.

the offer rate is 44.0 percent in the top search quintile and 25.0 percent in the bottom quintile. Workers with offers are slightly less likely to be unemployed in the next wave.

The middle quintiles are not perfectly monotone, so I treat this evidence as most informative at the distribution extremes. The purpose of the table is not to match the full cross-sectional distribution of search and offers. Rather, it checks whether the main empirical links behind the mechanism appear in held-out moments.

The calibrated model maps these validation moments into a simple flow mechanism:

$$p_{EU}(z) = \delta(z)(1 - p_{\text{offer}}(z, s(z))).$$

Holding fixed separation risk, higher offer receipt lowers the probability that a separation translates into unemployment. In recessions, separation risk rises, but workers also search more. The rise in search increases offer receipt and partially offsets the increase in unemployment inflows. Section 5 quantifies this offset.

The model is not designed to match the full cross-sectional dispersion in offer receipt. It uses a common offer technology and abstracts from worker heterogeneity in networks, occupations, search efficiency, and outside options. For this reason, the model should be interpreted as matching average offer rates and the behavioral response to perceived risk, rather than the entire distribution of offer receipt across workers.

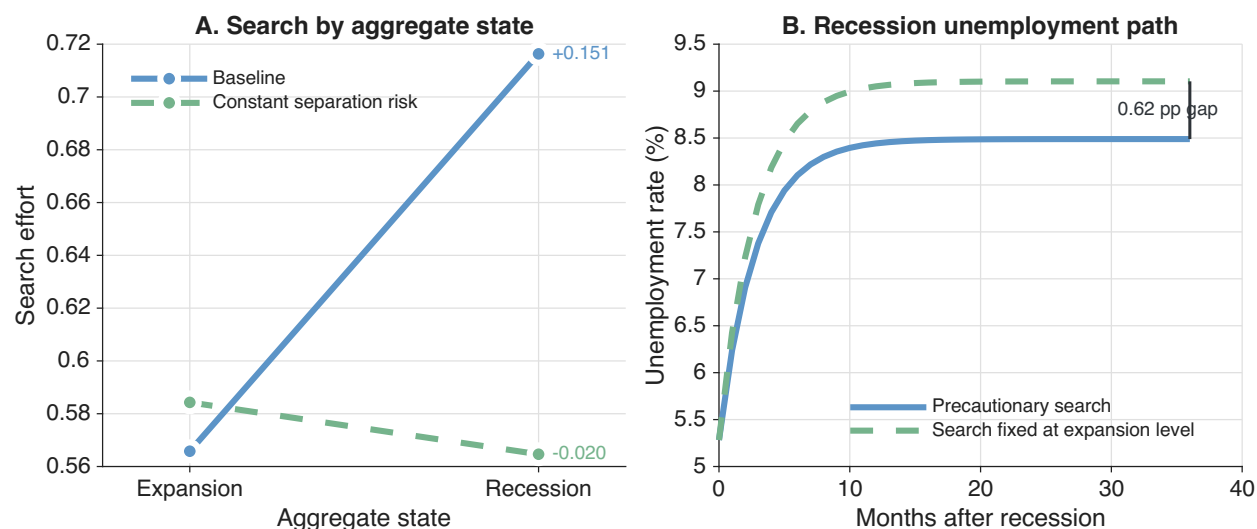
The untargeted moments line up with the model’s main links: perceived risk is higher among workers who search more, search is associated with more offer receipt, and offer receipt is associated with lower subsequent unemployment risk. The quantitative exercises below ask how much this margin matters for aggregate unemployment dynamics.

5 Aggregate Implications of Precautionary Search

The calibrated model matches the empirical relationship between perceived job-loss risk and search intensity. It also reproduces, without targeting them, the qualitative patterns in

offer receipt and subsequent unemployment reported in Section 4. I first summarize how search changes the recession unemployment path, then isolate the response to a shock in perceived job-loss risk holding actual labor-market conditions fixed. I then return to recession-state transitions and allow some workers to receive private information about potential job loss before separation occurs. The last exercises explain why search rises in recessions and why the unemployment effect depends on offers helping workers avoid unemployment.

Figure 3 summarizes the main quantitative mechanism. The left panel shows that search rises from expansion to recession in the baseline, while holding separation risk fixed reverses the slope. The right panel shows the aggregate implication on the natural unemployment-rate scale: precautionary search lowers the recession unemployment path relative to an economy in which employed search is fixed at its expansion level.



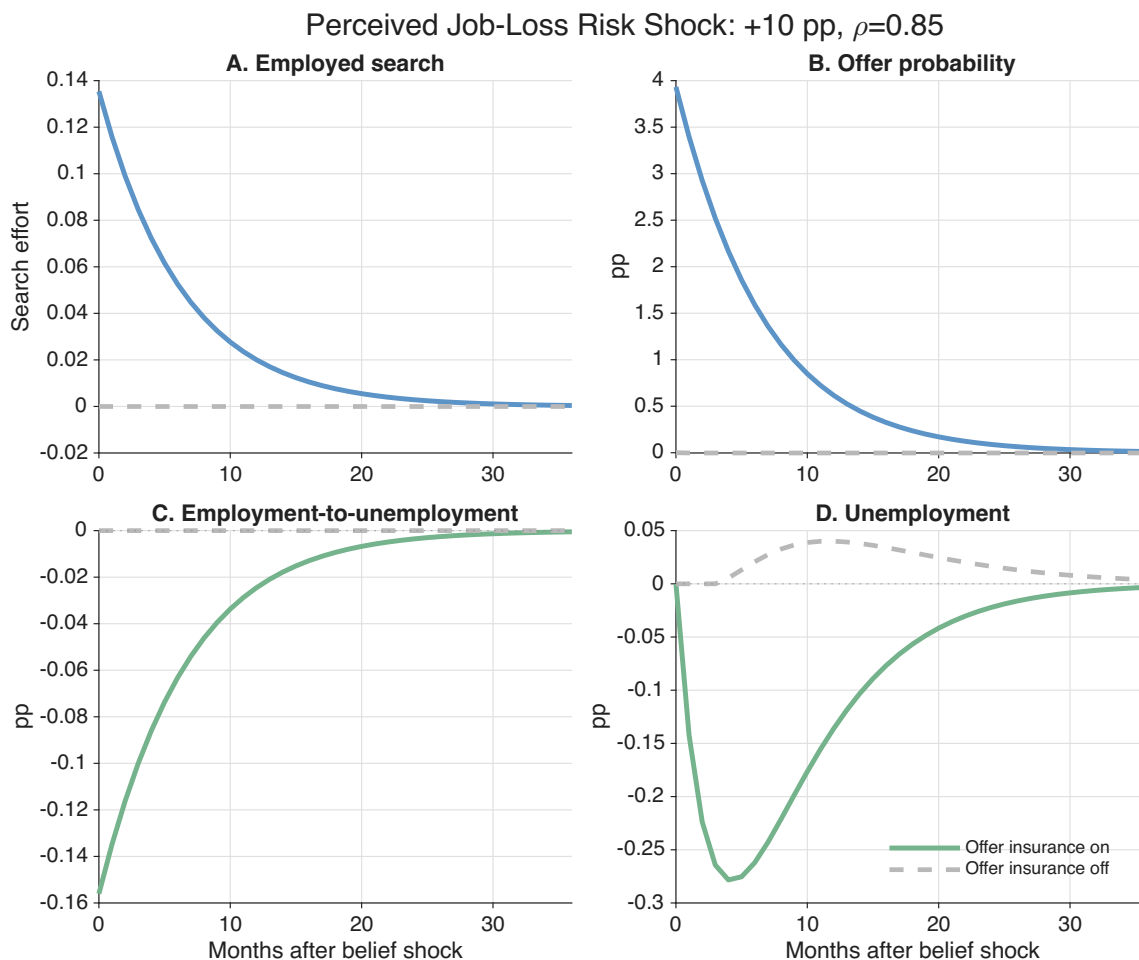
Notes: Panel A compares search effort across aggregate states in the baseline and in a counterfactual with constant separation risk. Panel B compares unemployment dynamics after a transition from the expansion state to the recession state. The fixed-search path holds employed search at its expansion level. The long-run gap is 0.62 percentage points.

Figure 3. Precautionary Search and Recession Unemployment

Figure 4 isolates the belief margin in the baseline model. I hold the aggregate state and objective separation risk fixed, and shock perceived 12-month job-loss risk by 10 percentage points. The shock follows an AR(1) process with monthly persistence $\rho = 0.85$. This exercise is not a recession shock: actual separation risk does not move. Instead, it asks how workers respond when they perceive higher job-loss risk, holding labor-market conditions fixed.

The response follows the mechanism in the model. Higher perceived job-loss risk raises employed search and the offer probability. Because objective separations are unchanged,

the higher offer probability lowers the employment-to-unemployment probability by increasing the share of separations covered by offers. The unemployment response is therefore negative and gradual. When offer insurance is shut down, the same belief shock has little effect on unemployment because offers no longer change whether separations become unemployment spells.

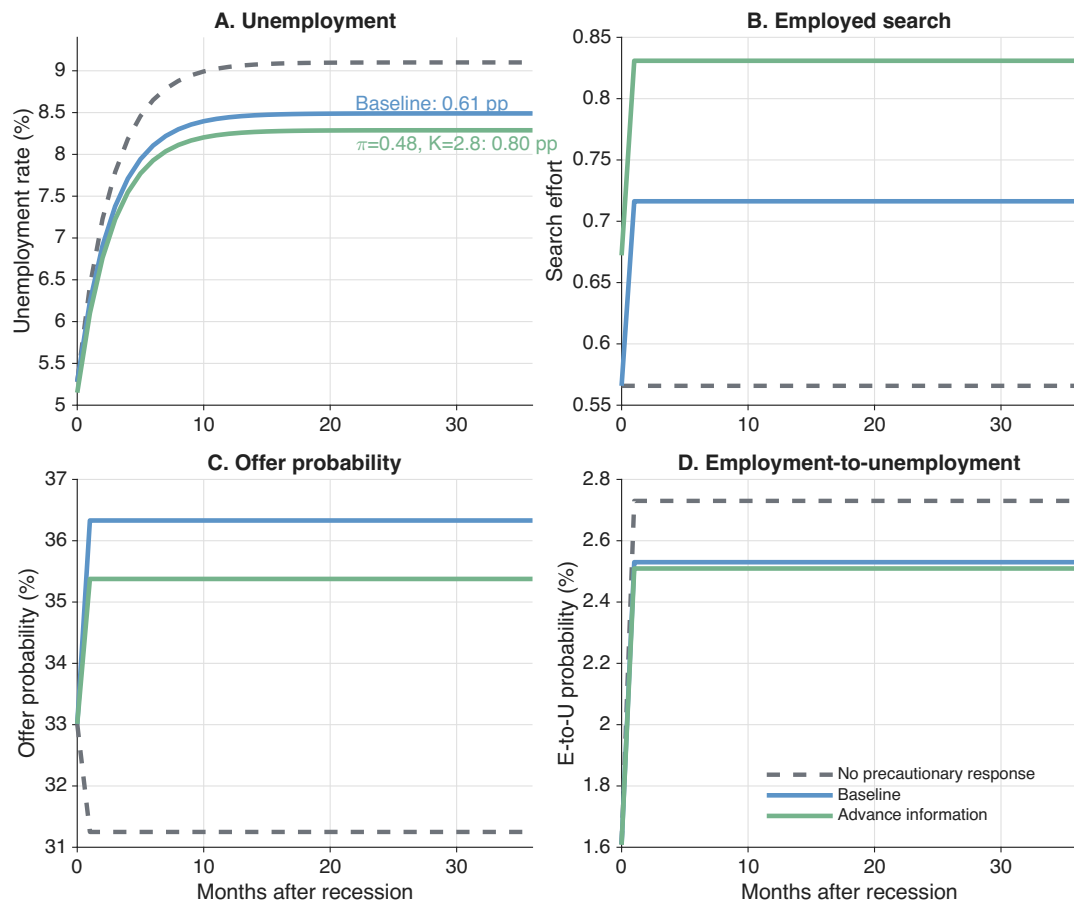


Notes: The figure reports responses to a 10 percentage point increase in perceived 12-month job-loss risk with monthly persistence $\rho = 0.85$. The aggregate state and objective separation risk are held fixed. The solid line keeps the offer-insurance mechanism active; the dashed line shuts down the mapping from offers to unemployment avoidance. Outcomes are deviations from the no-shock path.

Figure 4. Impulse Response to Perceived Job-Loss Risk

I next return to recession-state transitions and relax the baseline timing assumption. Figure 5 reports the advance-information extension introduced in Section 3.1. The baseline treats an outside offer as insurance only at the moment of separation. The extension calibrates the advance-information parameters from the SCE job supplement: among E-E transitions due to job loss or anticipated layoff, 47.5 percent report advance notice, and the average notice period among those with notice is 2.77 months. The figure shows

the unemployment path, employed search, the offer probability, and the employment-to-unemployment probability. Under the full recalibration, allowing workers to act on private information about potential job loss raises the long-run unemployment gap from 0.62 to 0.80 percentage points.



Notes: The figure plots model outcomes after a transition from the expansion state to the recession state. The no-precautionary-response path holds employed search fixed at its expansion level. The advance-information path uses $\pi = 0.475$ and $K = 2.77$, calibrated from the SCE job supplement, so some workers receive signals or private information about potential job loss before separation occurs. For this extension, the model is fully recalibrated after updating the separation target implied by the advance-information mapping.

Figure 5. Aggregate Implications Under Advance Information

5.1 Why Is On-the-Job Search Countercyclical?

In the calibrated equilibrium, on-the-job search is countercyclical: $s(B) > s(G)$. Table 10 reports search intensity by aggregate state. Workers search 27 percent more in recessions than in expansions ($\Delta s = 0.151$). This is the model counterpart of the empirical finding that workers with higher perceived job-loss risk search more while employed.

To isolate the role of cyclical separation risk, I consider a counterfactual in which the separation hazard is held constant across states, $\sigma(G) = \sigma(B) = \bar{\sigma}$, while all other mechanisms remain intact. In this economy, search becomes *procyclical* ($\Delta s = -0.020$). Without cyclical separation risk, lower market tightness in recessions reduces the return to search and workers search less. Appendix Table A15 reports the same sign reversal in an additional model run and includes an insurance-off benchmark.

Table 10. Search Cyclical Decomposition

Scenario	$s(G)$	$s(B)$	$\Delta s = s(B) - s(G)$
Baseline	0.566	0.716	+0.151
Constant σ	0.584	0.565	-0.020
Precautionary component			+0.170

Notes: Baseline: $\sigma(G) = 0.024$, $\sigma(B) = 0.040$. Constant- σ : $\bar{\sigma} = 0.026$ (ergodic mean). Positive Δs indicates countercyclical search.

When separation risk is constant, the only state dependence in the return to search operates through market tightness $\theta(z)$ and the employment surplus $W(z) - U(z)$. Because tightness declines in recessions, the offer arrival rate falls: even if a worker exerts effort, the probability of receiving an offer is lower when fewer vacancies are posted per unemployed worker. This decline in the return to search weakens incentives and renders search procyclical. Introducing cyclical separation risk reverses this logic. In a recession, workers face higher odds of losing their job, and an outside offer becomes more valuable. In the calibration, this increase in the value of search offsets the decline in tightness.

Decomposing the marginal benefit. To quantify the contribution of each force, I decompose the marginal benefit of search. The first-order condition implies

$$c'(s(z)) \propto \underbrace{\tilde{\delta}(z)}_{\text{Risk}} \cdot \underbrace{\frac{\partial p_{\text{offer}}(z, s(z))}{\partial s}}_{\text{Offer response}} \cdot \underbrace{\sum_{z'} \Pi(z, z') (W(z') - U(z'))}_{\text{Expected surplus}}.$$

Table 11 reports counterfactual search differences when each component is held at its ergodic mean. Appendix Table A16 reports the corresponding marginal-benefit index from the additional model run.

The table shows that tightness and surplus both work *against* countercyclical search: holding either constant increases Δs . The precautionary motive alone would generate $\Delta s = 0.192$, but two countervailing forces partially offset this effect.

First, procyclical tightness ($\lambda_e(B) < \lambda_e(G)$) reduces the return to search in recessions.

Table 11. Marginal Benefit Decomposition

Counterfactual	$\Delta s = s(B) - s(G)$	Direction
Baseline	+0.124	Countercyclical
Constant perceived risk ($\tilde{\sigma}$)	-0.141	Procyclical
Constant offer response (λ_e)	+0.147	More countercyclical
Constant surplus ($W - U$)	+0.167	More countercyclical
Risk only (both λ_e and $W - U$ fixed)	+0.192	Strongest

Notes: Each row holds one component at its ergodic mean while allowing others to vary. “Risk only” holds both tightness and surplus fixed, isolating the precautionary channel.

When labor markets are slack, vacancies are scarce relative to job seekers, and the probability of receiving an offer falls even for workers who search intensively. Second, procyclical surplus ($(W - U)(B) < (W - U)(G)$) reduces the value of an offer when one arrives. In bad times, both employment and unemployment become less attractive, but the gap between them narrows because the option value of search from unemployment declines. These forces attenuate the search response to higher separation risk, yielding the observed $\Delta s = 0.151$ rather than the larger response that would occur if only separation risk varied.

This exercise points to the same conclusion as the fixed-separation counterfactual. Cyclical perceived risk is the main reason search rises in recessions in the calibrated model. Without it, workers would reduce search effort in recessions. The other forces, tightness and surplus, work in the opposite direction but are quantitatively weaker.

5.2 Precautionary Search Dampens Recessions

The previous exercise shows that on-the-job search rises in recessions. I next compare the baseline recession steady state to a counterfactual in which search is held fixed at its expansion level. The counterfactual asks: if workers did not respond to heightened separation risk by searching more, how much higher would unemployment rise?

Table 12 reports the comparison.

When a recession occurs, workers increase search from 0.57 to 0.72, a 27 percent rise in effort. The higher effort raises the offer probability from 0.33 to 0.36, meaning that more workers hold outside options when separation risk is elevated. This reduces the fraction of separations that translate into unemployment: a worker who would otherwise enter the unemployment pool instead transitions directly to a new job.

If search were held fixed at its expansion level, unemployment would rise by 3.82 percentage points instead of 3.21 percentage points when search responds endogenously. The difference, 0.62 percentage points, represents 16.1 percent of the total recessionary

Table 12. Recession Steady States: Baseline vs. Fixed Search

	s	p_{offer}	u	Δu
G steady state	0.566	0.330	5.28%	—
B steady state (baseline)	0.716	0.363	8.49%	+3.21 pp
B steady state (fixed s)	0.566	0.313	9.10%	+3.82 pp
Dampening	0.62 pp (16.1%)			

Notes: Baseline allows search to re-optimize; fixed- s holds search at $s(G)$. Dampening = $(3.82 - 3.21)/3.82$.

increase in unemployment. Put differently, about one-sixth of the rise in unemployment that would occur with fixed employed search is avoided because workers search more when job-loss risk is higher.

The benchmark keeps $m(G) = m(B) = 1$, so perceived risk inherits its cyclical movement from actual separation risk. Table 13 adds the macro-to-personal belief movement estimated in the SCE. I set $m(G) = 1$ and choose $m(B) = 1.17$ so that the recession state adds 4.05 percentage points to annual perceived job-loss risk, equal to the first-stage coefficient in Table 3 multiplied by a one-standard-deviation increase in macro pessimism. This version keeps the same mean perceived job-loss target and recalibrates the search block to the same offer and search-response moments.

Table 13. Benchmark and Recession-Belief Calibration

	Benchmark	Recession-belief version
$m(G), m(B)$	1.00, 1.00	1.00, 1.17
ζ	0.775	0.759
Annual perceived job-loss risk, G	20.2%	19.9%
Annual perceived job-loss risk, B	31.4%	35.1%
Search increase, G to B	27%	38%
Offer probability, B	0.363	0.382
Unemployment, B	8.49%	8.26%
Fixed-search unemployment, B	9.10%	9.10%
Dampening	0.62 pp (16.1%)	0.84 pp (22.1%)

Notes: The benchmark sets $m(G) = m(B) = 1$, so perceived risk moves with actual separation risk. The recession-belief version uses the SCE macro-to-personal belief estimate to let perceived job-loss risk rise more in the recession state. Both versions match the mean perceived job-loss probability, the employed offer probability, and the search response to perceived job-loss risk.

Table 14 reports the robustness checks requested by the model audit. I vary the estimated search response, the share of offers that can help a worker avoid unemployment, the offer target, the cyclical of separation risk, unemployment value b , matching elasticity

η , and search-cost curvature. The effect remains meaningful in most cases where offers can be used to avoid unemployment. It becomes much smaller when fewer offers can play that role, which is exactly what the mechanism predicts.

Table 14. Sensitivity of Recession Dampening to Model Inputs

Scenario	Search increase	$p_{offer}(B)$	Dampening	Share
Baseline	26.6%	0.363	0.62 pp	16.1%
$\hat{\beta}_{loss} = 25\%$ of baseline	30.4%	0.381	0.84 pp	21.9%
$\hat{\beta}_{loss} = 50\%$ of baseline	28.8%	0.374	0.74 pp	19.5%
$\hat{\beta}_{loss} = 75\%$ of baseline	27.5%	0.368	0.67 pp	17.6%
$\hat{\beta}_{loss} = 100\%$ of baseline	26.6%	0.363	0.62 pp	16.1%
Usable-offer share $\rho_I = 0.25$	28.1%	0.371	0.12 pp	3.4%
Usable-offer share $\rho_I = 0.50$	27.6%	0.369	0.26 pp	7.3%
Usable-offer share $\rho_I = 0.75$	27.2%	0.366	0.42 pp	11.5%
Usable-offer share $\rho_I = 1.00$	26.6%	0.363	0.62 pp	16.1%
Lower offer target: $p_{offer}(G) = 0.25$	28.6%	0.282	0.49 pp	13.1%
Separation ratio $\sigma(B)/\sigma(G) = 1.30$	11.3%	0.340	0.23 pp	12.0%
Separation ratio $\sigma(B)/\sigma(G) = 2.00$	38.7%	0.379	0.98 pp	18.0%
$b = 0.30$	27.3%	0.366	0.59 pp	16.8%
$b = 0.50$	25.7%	0.359	0.64 pp	15.2%
$\eta = 0.05$	26.6%	0.363	0.62 pp	16.1%
$\eta = 0.20$	26.6%	0.363	0.62 pp	16.1%
Search-cost curvature $\nu = 0.75$	32.4%	0.375	0.76 pp	19.9%
Search-cost curvature $\nu = 1.50$	19.7%	0.349	0.44 pp	11.5%

Notes: Each row recalibrates the model after changing the listed input and reports the recession steady-state comparison between endogenous search and search fixed at its expansion value. Matching parameters and the baseline belief scale are calibrated once and then held fixed across scenario rows. “Dampening” is $100 \times (u_B^{fixed} - u_B)$ in percentage points. “Share” divides this dampening by the recessionary unemployment increase under fixed search. The $\hat{\beta}_{loss}$ rows scale the empirical search response used to discipline search costs. The ρ_I rows treat only a fraction of offers as usable for avoiding unemployment after separation. Failed rows, if any, indicate that the nested calibration could not hit the required targets under that scenario.

To put this magnitude in perspective, the 0.62 percentage point attenuation is comparable to estimates of the stabilizing effect of automatic unemployment insurance extensions during recessions.¹² The magnitude is also similar to [Mukoyama et al. \(2018\)](#), who find that countercyclical search effort among the *unemployed* reduced unemployment by 0.5–1 percentage point during the Great Recession. My estimate complements theirs: they study unemployed search while I focus on employed search, yet both margins dampen unemployment fluctuations through search behavior.

The mechanism differs from UI extensions, however. UI extensions operate through the outflow margin: they provide income support that allows unemployed workers to search more effectively, speeding their return to employment. Precautionary on-the-job

¹²For estimates of the stabilizing effect of UI extensions, see [Hagedorn et al. \(2013\)](#) and [Chodorow-Reich and Karabarbounis \(2019\)](#), who find effects in the range of 0.3–0.8 percentage points depending on specification and time period.

search operates through the inflow margin: it reduces the number of separations that result in unemployment in the first place. Unlike UI extensions, precautionary search operates automatically through individual optimization and requires no policy intervention. Workers respond to higher perceived risk because it is privately optimal to do so, and this private response generates a positive externality for aggregate stability.

Mechanism. The dampening operates entirely through the inflow margin. Higher search increases the probability of holding an outside offer at the time of separation, lowering the separation-to-unemployment probability δ_{EU} . Consider a worker who is separated in a recession. In the baseline economy, where search has responded to heightened risk, this worker is more likely to hold an offer and can avoid an unemployment spell. In the fixed-search counterfactual, offer holdings are lower, and the same separation is more likely to result in joblessness. The dampening effect arises entirely from this margin: fewer separations resulting in unemployment spells rather than faster exits from unemployment.

5.3 Do Offers Have to Prevent Unemployment?

The fixed-search exercise shuts down the search response entirely. A complementary exercise keeps search active but changes what offers do. It asks whether higher search still lowers unemployment when offers no longer let separated workers avoid unemployment.

In the baseline economy, an employed worker who receives an offer prior to separation can move directly to a new job and avoid unemployment. In the “insurance-off” counterfactual, separated workers enter unemployment regardless of whether they hold an offer.

The key question is whether the search response still lowers unemployment. If the mechanism is really about offers helping workers avoid unemployment, then the baseline economy should show lower unemployment inflows while the insurance-off economy should not.

To implement this comparison, I consider a transitory increase in perceived job-loss risk of 10 percentage points, with persistence $\rho = 0.85$, starting from the recession state.¹³ This shock mimics the within-period variation in beliefs documented in the empirical analysis, where workers who perceive higher personal risk search more intensively. Table 15 reports the response.

¹³The persistence $\rho = 0.85$ implies a half-life of approximately 4 months, consistent with the autocorrelation of perceived job-loss risk in the SCE panel.

Table 15. Offers and Unemployment Avoidance

	Insurance On	Insurance Off
δ_{EU} at impact	0.0253	0.0397
Δu at impact	-0.13 pp	0.00 pp
Offer-coverage share	36.3%	

Notes: Belief shock: 10 pp increase in perceived job-loss probability, $\rho = 0.85$. Offer-coverage share = $(\delta_{EU}^{\text{off}} - \delta_{EU}^{\text{on}}) / \delta_{EU}^{\text{off}}$.

The difference lies in what happens after a worker searches more. When offers can prevent unemployment, higher search lowers δ_{EU} by 36 percent relative to the no-insurance economy. More workers hold offers at the moment of separation, and those offers keep them out of unemployment. When offers do not prevent unemployment, separations determine unemployment inflows, and the belief shock has no impact effect on unemployment. Appendix Table A17 reports the full dynamic path.

This exercise should not be read as saying that one part of the recession effect comes from offers and the rest comes from firm responses. Instead, it shows a simpler point. Search matters for unemployment because it raises the chance that a worker has an offer when job loss occurs. Market tightness and the value of employment still shape how much workers search, but the unemployment effect comes from fewer separations becoming unemployment spells.

This point is useful for thinking about policy. Policies that make it easier for employed workers to find credible outside offers could strengthen this stabilizing margin. Policies that discourage on-the-job search could weaken it.

5.4 Discussion

The counterfactuals show how the search response affects aggregate unemployment. Countercyclical search in the model is not automatic: it emerges because job-loss risk rises in recessions enough to offset weaker labor-market conditions. When separation risk is held constant, search becomes procyclical. Endogenous precautionary search reduces the recessionary rise in unemployment by about 16 percent. The reason is that higher search increases the chance of having an offer when separation risk is high, so fewer separations become unemployment spells.

These findings do not rely on belief distortions or state-dependent optimism. In the baseline calibration, perceived separation risk inherits its cyclicity entirely from the objective separation hazard. The mechanism requires only that workers respond to higher

perceived risk by searching more intensively, the behavioral channel identified in the empirical analysis and captured by the positive semi-elasticity $\hat{\beta}_{\text{plse}}$. The empirical semi-elasticity enters the model through calibration, and the model traces out its aggregate implications.

Caveats. Several limitations should inform interpretation of the quantitative results. The model treats separations as exogenous and abstracts from endogenous quits that respond differently to aggregate conditions. In practice, some workers who perceive high separation risk may quit preemptively if they secure a suitable offer; this margin is not modeled. Match quality is homogeneous, so the framework cannot address whether precautionary search leads workers into worse matches or generates inefficient reallocation. If workers accept lower-quality matches to avoid unemployment, the welfare implications of precautionary search could be more nuanced than the unemployment statistics suggest.

The analysis focuses on the employed search margin and does not model potential discouragement effects among the unemployed. If unemployed workers reduce search effort during recessions, this could offset or reinforce the stabilization from employed search depending on relative magnitudes. The model also abstracts from heterogeneity in search costs, time preferences, and asset holdings that could generate rich distributional effects. Finally, the quantitative results rest on cross-sectional identification of the expectations-search relationship. If unobserved heterogeneity drives both pessimism and search propensity, the estimated semi-elasticity may overstate the causal effect. The goal is to show that a calibration disciplined by microdata implies a nontrivial aggregate effect.

6 Conclusion

This paper shows that employed workers who perceive higher job-loss risk search more while employed. In the Survey of Consumer Expectations, reported job-loss probabilities predict search intensity, forecast subsequent separations, and are linked to security-related search motives. The response is strongest when workers perceive both high job-loss risk and weak re-employment prospects, consistent with a precautionary motive.

In the calibrated model, workers search about 27 percent more in recessions, and that response reduces the recessionary rise in unemployment by about 16 percent. The mechanism is simple: higher search makes workers more likely to have an outside offer when job loss occurs, and those offers help some workers avoid unemployment. When offers are not allowed to play this role in the model, belief-induced search no longer lowers unemployment. The key caveat is identification: these magnitudes use cross-sectional

evidence on beliefs and search, but sharper causal variation in beliefs would be needed for stronger policy conclusions.

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A Data Appendix: Survey of Consumer Expectations (SCE)

This appendix provides additional detail on the Survey of Consumer Expectations (SCE) and the construction of the empirical analysis sample. The main text documents that perceived job-loss risk predicts on-the-job search intensity and motivates a precautionary interpretation. Because this interpretation hinges on how expectations and search are measured, this appendix clarifies survey design, variable construction, and sample selection.

A.1 Survey design and panel structure

The SCE is a nationally representative monthly internet panel administered by the Federal Reserve Bank of New York since 2013. Approximately 1,300 household heads participate each month. Respondents remain in the panel for up to 12 consecutive months, allowing expectations and realized outcomes to be linked within person. All reported statistics use SCE survey weights. Regressions cluster standard errors at the individual level to account for within-person serial correlation.

The panel structure is central for two reasons emphasized in the main text. First, it allows validation of beliefs against subsequent realizations (Section 2.4). Second, it permits specifications using lagged beliefs and within-person variation, reported below, to address concerns about contemporaneous shocks jointly driving expectations and search.

A.2 Core survey and labor-market expectations

The monthly *core survey* elicits probabilistic expectations on a 0–100 scale. The analysis uses four measures:

1. *Macroeconomic pessimism*: the subjective probability that the U.S. unemployment rate will be higher in 12 months.
2. *Job-loss probability*: the subjective probability of involuntarily losing the current job within 12 months.
3. *Voluntary separation probability*: the subjective probability of leaving the job voluntarily within 12 months.
4. *Job-finding probability conditional on job loss*: the subjective probability of finding a job within three months if the current job is lost.

All expectations are rescaled to $[0, 1]$. The distinction between involuntary job-loss beliefs and voluntary separation beliefs is economically important. The model’s precautionary motive operates through perceived *inflow risk* into unemployment. The empirical analysis therefore distinguishes involuntary risk from voluntary mobility and from beliefs about re-employment prospects.

A.3 Labor Market Module and search measurement

The *Labor Market Module* (LMM) is administered every four months to respondents while in the panel. The primary outcome variable is weekly job-search intensity: total hours spent searching or applying for jobs in the past seven days. For each LMM interview month, I match expectations from the core survey in the same calendar month (or the most recent prior core-survey response if the same-month observation is missing).

Search hours are winsorized at the 5th and 95th percentiles to limit the influence of extreme reports. The right-skewed distribution motivates robustness checks using alternative functional forms (Appendix Table A5). Importantly, the model interprets search as an intensity choice, not literal hours; accordingly, the calibration targets the *semi-elasticity* of search with respect to perceived job-loss risk rather than the level of hours.

The LMM also records industry of current employment (LMind). This variable is used to construct industry fixed effects and industry×time fixed effects in robustness specifications reported below. These specifications address the concern that unobserved industry-level shocks may jointly affect beliefs and search.

A.4 Appendix Table A1: Summary Statistics

Appendix Table A1 reports weighted summary statistics for the employed LMM sample.

Two features are worth emphasizing. First, subjective expectations display substantial dispersion. The cross-sectional standard deviation of job-loss beliefs is comparable to its mean. This dispersion underlies the main empirical design: within a given state and month, workers hold different beliefs about separation risk.

Second, search intensity is highly skewed. The mean is approximately 3.8 hours per week, but the standard deviation exceeds the mean. This motivates both winsorization and the use of alternative specifications in Appendix Table A5. The empirical results in the main text do not depend on functional form.

Table A1. Summary Statistics: Employed Workers

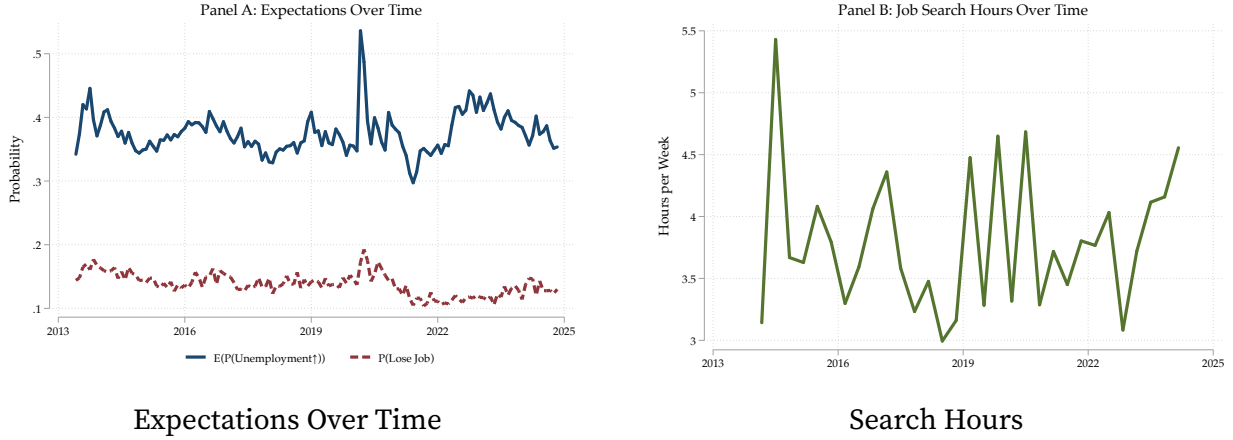
	N	Mean	SD
Search hours (winsorized)	4226	3.774	4.800
$E_t(\Delta \Pr(U_t + 12 \geq 0))$	4217	0.394	0.239
$E_t(\Pr(\text{Lose Job}_t + 12))$	4220	0.213	0.234
$E_t(\Pr(\text{Leave Job}_t + 12))$	4220	0.402	0.304
$E_t(\Pr(\text{Find} \text{Lose}_t + 3))$	4219	0.571	0.300
$E_t(\pi_t + 12)$ (point)	4219	0.072	0.213
Age	4226	41.508	11.911
Female	4224	0.453	0.498
Married	4225	0.599	0.490
White	4226	0.772	0.419
Black	4226	0.145	0.352
Hispanic	4225	0.120	0.325
HH income: Under 50k	4211	0.376	0.484
HH income: 50k–100k	4211	0.324	0.468
HH income: Over 100k	4211	0.300	0.459
Education: High school	4205	0.240	0.427
Education: Some college	4205	0.293	0.455
Education: College+	4205	0.467	0.499

Notes: Sample: employed respondents in the SCE Labor Market Module with non-missing search hours (2014–2023). Search hours winsorized at the 5th/95th percentiles. Expectations scaled to $[0, 1]$. Means/SD weighted by SCE analytic weights; N is unweighted non-missing count. Binary indicators report sample shares.

A.5 Appendix Figure A1: Aggregate Time Series

B Empirical Appendix

This section collects robustness and identification checks referenced in Section 2. Each exhibit addresses a specific concern about the interpretation of the belief–search relationship. Throughout, the central question is whether the positive association between perceived job-loss risk and on-the-job search intensity reflects a precautionary response to perceived separation risk, or instead captures omitted local shocks, sectoral conditions, functional-form artifacts, or stable individual heterogeneity.



Notes: The figure plots weighted monthly means of macro unemployment expectations (core survey) and weekly search hours (LMM) for employed respondents. Expectations spike during major aggregate events (notably COVID-19). Search hours are more volatile because the LMM is only administered every four months to a rotating subsample. The aggregate time series is therefore noisy, motivating the use of within-state-month cross-sectional variation in the main analysis.

Figure A1. Time Series of Expectations and Search Intensity

B.1 Alternative Fixed Effects

The specification replaces separate state and time fixed effects with their full interaction:

$$\text{SearchHours}_{ist} = \beta' \mathbf{E}_{it} + \mathbf{X}_{it}' \gamma + \mu_{st} + \varepsilon_{ist}, \quad (29)$$

where $\mathbf{E}_{it}$ collects the expectation variables (added progressively across columns), $\mathbf{X}_{it}$ are demographic controls, and μ_{st} denotes state×time fixed effects. Standard errors are clustered at the individual level; observations are weighted by survey weights. The sample consists of employed workers.

State×time fixed effects. This specification absorbs all state-level labor-market conditions in a given month, including unemployment rates, vacancy conditions, and other local shocks. Identification therefore comes entirely from cross-sectional differences in beliefs among workers residing in the *same state in the same month*.

The stability of the job-loss coefficient relative to the baseline indicates that the belief–search relationship is not driven by common local shocks. Even within a fixed local environment, workers who perceive higher separation risk search more intensively.

This specification replaces time fixed effects with industry×time interactions and retains state fixed effects:

$$\text{SearchHours}_{ijst} = \beta' \mathbf{E}_{it} + \mathbf{X}_{it}' \gamma + \mu_s + \delta_{jt} + \varepsilon_{ijst},$$

Table A2. Expectations and On-the-Job Search Intensity

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
$E_t[\Delta \Pr(U_{t+12} \geq 0)]$	1.538*** (0.420)	0.920** (0.415)	0.906** (0.412)	0.814* (0.435)	0.870** (0.408)
$E_t \Pr(\text{Lose Job}_{t+12})$		2.064*** (0.499)	1.841*** (0.502)	1.533*** (0.517)	2.719*** (0.991)
$E_t \Pr(\text{Leave Job}_{t+12})$		1.372*** (0.398)	1.691*** (0.407)	1.040** (0.439)	1.692*** (0.407)
$E_t \Pr(\text{Find Job}_{t+3})$			-1.259*** (0.352)	-1.762*** (0.360)	-0.973** (0.441)
$E_t \Pr(\text{Offer}_{t+3})$				0.020*** (0.004)	
$E_t(\Pr(\text{Lose Job}_{t+12})) \times$ $E_t(\Pr(\text{Find Job}_{t+3}))$					-1.523 (1.452)
$E_t(\pi_{t+12})$	-0.019 (0.023)	-0.019 (0.023)	-0.023 (0.023)	-0.019 (0.024)	-0.024 (0.023)
Separation expectations	No	Yes	Yes P(find)	Yes	Yes
Job-finding expectations	No	No	given lose	Both	Interac- tion
State $\times$ Time FE	Yes	Yes	Yes	Yes	Yes
Observations	4132	4131	4130	3922	4130
R-squared	0.391	0.405	0.409	0.419	0.409

Notes: Dependent variable: weekly search hours (winsorized). Expectations scaled to $[0, 1]$. Specification absorbs state $\times$ time fixed effects; controls include age polynomial, education, race, gender, marital status, household income; weighted by survey weights; standard errors clustered by individual; sample: employed workers. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

where δ_{jt} denotes industry $\times$ time fixed effects (using the LMM industry classification) and μ_s are state fixed effects.

Industry $\times$ time fixed effects. Industry $\times$ time fixed effects absorb sector-specific shocks at a monthly frequency, addressing the concern that industry downturns jointly raise perceived risk and search.

The job-loss coefficient remains economically and statistically meaningful under this demanding specification. This indicates that the belief–search relationship is not merely reflecting industry-wide conditions, but also heterogeneity in perceived risk among workers in the same industry at the same point in time.

Table A3. Expectations and On-the-Job Search Intensity

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
E_t					
$\Delta \Pr(U_{t+12} \geq 0)$	1.671*** (0.452)	0.981** (0.435)	1.010** (0.435)	1.110** (0.454)	1.006** (0.432)
E_t					
$\Pr(\text{Lose Job}_{t+12})$		2.107*** (0.483)	1.844*** (0.493)	1.652*** (0.515)	1.976** (0.975)
E_t					
$\Pr(\text{Leave Job}_{t+12})$		1.595*** (0.369)	1.917*** (0.379)	1.411*** (0.415)	1.917*** (0.379)
E_t					
$\Pr(\text{Find Job}_{t+3})$			-1.222*** (0.342)	-1.566*** (0.358)	-1.176*** (0.441)
E_t					
$\Pr(\text{Offer}_{t+3})$				0.014*** (0.004)	
$E_t \Pr(\text{Lose Job}_{t+12})$ $\times E_t \Pr(\text{Find Job}_{t+3})$					-0.227 (1.432)
$E_t(\pi_{t+12})$ (dist. mean)	-0.014 (0.027)	-0.019 (0.027)	-0.023 (0.027)	-0.020 (0.029)	-0.023 (0.027)
Separation expectations	No	Yes		Yes	Yes
Job-finding expectations	No	No	P(find) given lose	Both	Interac- tion
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	3086	3085	3085	2925	3085
R-squared	0.332	0.350	0.354	0.361	0.354

Notes: Dependent variable: weekly search hours (winsorized). Expectations scaled to $[0, 1]$. Specification absorbs industry $\times$ time fixed effects and includes state FE; controls include age polynomial, education, race, gender, marital status, household income; weighted by survey weights; standard errors clustered by individual; sample: employed workers. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The sample is restricted to employed workers with non-missing industry. Column (1) uses state×time fixed effects (μ_{st}); Column (2) adds industry fixed effects (δ_j); Column (3) replaces these with industry×time fixed effects (δ_{jt}) alongside state fixed effects (μ_s).

Table A4. Main Result Robustness: Industry Controls (LMind)

	(1) State×Time FE	(2) + Industry FE	(3) Industry×Time FE
$E_t[\Delta \Pr(U_{t+12} \geq 0)]$	0.853* (0.499)	0.772 (0.508)	1.006** (0.432)
$E_t \Pr(\text{Lose Job}_{t+12})$	1.310 (1.152)	1.379 (1.161)	1.976** (0.975)
$E_t \Pr(\text{Leave Job}_{t+12})$	2.104*** (0.464)	2.068*** (0.464)	1.917*** (0.379)
$E_t \Pr(\text{Find Job}_{t+3})$	-1.433** (0.588)	-1.356** (0.575)	-1.176*** (0.441)
$E_t \Pr(\text{Lose Job}_{t+12}) \times$ $E_t \Pr(\text{Find Job}_{t+3})$	0.466 (1.775)	0.460 (1.769)	-0.227 (1.432)
$E_t(\pi_{t+12})$	-0.017 (0.029)	-0.014 (0.029)	-0.023 (0.027)
State × Time FE	Yes	Yes	No
Industry × Time FE	No	No	Yes
Industry FE	No	Yes	No
Observations	3085	3085	3085
R-squared	0.489	0.495	0.354

Notes: Sample restricted to employed workers with non-missing LMind industry. Column 1 absorbs state×time FE; column 2 adds industry FE; column 3 absorbs industry×time FE and includes state FE. Expectations scaled to $[0, 1]$; weighted by survey weights; standard errors clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Industry and state controls combined. Adding industry fixed effects alongside time and state controls further isolates within-industry, within-state belief variation. The magnitude and sign of the key coefficients are preserved. Taken together, these specifications show that the precautionary relationship survives controls for both geographic and sectoral conditions.

Importantly, these exercises do not establish causality. They show that the empirical relationship is not mechanically driven by observable local or sectoral shocks, strengthening the case that belief heterogeneity plays an independent role.

B.2 Alternative Outcomes and Timing

Columns (1)–(3) estimate the baseline specification with alternative dependent variables:

$$f(\text{SearchHours}_{ist}) = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it} \gamma + \mu_s + \lambda_t + \varepsilon_{ist},$$

where $f(\cdot)$ is the identity (levels), $\log(1 + \cdot)$, or the inverse hyperbolic sine. Column (4) estimates a Poisson quasi-maximum likelihood model with the same regressors and fixed effects.

Table A5. Robustness: Alternative Dependent Variable Specifications

	(1) Levels	(2) Log	(3) Arc-%	(4) Poisson
main				
$E_t(\Delta \Pr(U_t + 12 \geq 0))$	0.953*** (0.355)	0.130** (0.058)	0.144** (0.070)	0.246*** (0.091)
$E_t(\Pr(\text{Lose Job}_t + 12))$	1.836*** (0.429)	0.344*** (0.065)	0.413*** (0.077)	0.398*** (0.094)
$E_t(\Pr(\text{Leave Job}_t + 12))$	1.884*** (0.333)	0.389*** (0.052)	0.479*** (0.062)	0.512*** (0.083)
$E_t(\Pr(\text{Find} \text{Lose}_t + 3))$	-1.156*** (0.294)	-0.210*** (0.050)	-0.251*** (0.060)	-0.351*** (0.078)
Dep. Variable	Levels	Log(1+y)	Arc-%	Poisson
Observations	4130	4130	4130	4130
R-squared	0.152	0.159	0.156	

Notes: Dependent variable variants: levels, $\log(1+y)$, arc-% $(y - \bar{y}) / (0.5(y + \bar{y}))$, and Poisson. Expectations scaled to $[0, 1]$. All specifications include state and time fixed effects plus controls; weighted by survey weights; standard errors clustered by individual (Poisson uses robust SE). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Functional form robustness. Search hours are highly skewed, with a substantial mass at zero. Appendix Table A5 shows that the positive association between perceived job-loss probability and search intensity is robust across alternative transformations and specifications, including $\log(1 + y)$, inverse hyperbolic sine, and Poisson quasi-maximum likelihood.

The dependent variable is an indicator for any positive search hours:

$$\mathbb{1}\{\text{SearchHours}_{ist} > 0\} = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it} \gamma + \mu_s + \lambda_t + \varepsilon_{ist}.$$

Table A6. Expectations and On-the-Job Search: Extensive Margin

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) + P(offer)	(5) Interaction
E_t $\Delta \Pr(U_{t+12} \geq 0)$	0.039 (0.026)	0.014 (0.026)	0.015 (0.026)	0.003 (0.026)	0.013 (0.026)
E_t $\Pr(\text{Lose Job}_{t+12})$		0.072*** (0.024)	0.066*** (0.024)	0.057** (0.025)	0.101** (0.051)
E_t $\Pr(\text{Leave Job}_{t+12})$		0.064*** (0.021)	0.073*** (0.022)	0.040 (0.024)	0.073*** (0.022)
E_t $\Pr(\text{Find Job}_{t+3})$			-0.033 (0.023)	-0.054** (0.023)	-0.022 (0.029)
E_t $\Pr(\text{Offer}_{t+3})$				0.001*** (0.000)	
$E_t \Pr(\text{Lose Job}_{t+12})$ $\times E_t \Pr(\text{Find Job}_{t+3})$					-0.062 (0.081)
$E_t(\pi_{t+12})$ (dist. mean)	-0.002 (0.001)	-0.002 (0.001)	-0.002 (0.001)	-0.002 (0.001)	-0.002 (0.001)
Separation expectations	No	Yes	Yes P(find)	Yes	Yes
Job-finding expectations	No	No	given lose	Both	Interac- tion
Time FE	Yes	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes	Yes
Observations	4132	4131	4130	3922	4130
R-squared	0.045	0.051	0.052	0.061	0.052

Notes: Dependent variable: indicator for any search (> 0 hours). Expectations scaled to $[0, 1]$. State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual; sample: employed workers. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Extensive margin. Appendix Table A6 shows that perceived job-loss probability significantly predicts the probability of engaging in any search at all. Macro pessimism loses statistical significance once personal beliefs are included, consistent with the separation-risk-first interpretation emphasized in the main text.

Contemporaneous expectations are replaced with their values from the most recent prior core-survey wave:

$$\text{SearchHours}_{ist} = \beta' \mathbf{E}_{i,t-1} + \mathbf{X}_{it}' \gamma + \mu_s + \lambda_t + \varepsilon_{ist},$$

where $\mathbf{E}_{i,t-1}$ denotes expectations measured before the LMM interview.

Table A7. Lagged Expectations and On-the-Job Search Intensity

	(1) Baseline	(2) + Separation	(3) + P(find)	(4) Interaction
Lagged $E_t(\Delta \Pr(U_t + 12 \geq 0))$	1.322*** (0.405)	0.453 (0.428)	0.419 (0.429)	0.384 (0.429)
Lagged $E_t(\Pr(\text{Lose Job}_t + 12))$		2.323*** (0.484)	2.097*** (0.490)	2.918*** (0.892)
Lagged $E_t(\Pr(\text{Leave Job}_t + 12))$		1.483*** (0.334)	1.751*** (0.336)	1.773*** (0.337)
Lagged $E_t(\Pr(\text{Find} \text{Lose}_t + 3))$			-1.201*** (0.292)	-0.948*** (0.336)
Lagged $E_t(\Pr(\text{Offer}_t + 3))$				
Lagged $E_t(\Pr(\text{Lose Job}_t + 12)) \times$ Lagged $E_t(\Pr(\text{Find} \text{Lose}_t + 3))$				-1.444 (1.336)
Lagged $E_t(\pi_t + 12)$ (dist. mean)	0.014 (0.021)	0.020 (0.022)	0.018 (0.022)	0.018 (0.022)
Separation expectations	No	Yes	Yes P(find)	Yes
Job-finding expectations	No	No	given lose	Interac- tion
Time FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Observations	4133	3975	3974	3974
R-squared	0.122	0.139	0.144	0.145

Notes: Lagged expectations from the prior core survey wave (≈ 4 months earlier). Dependent variable: weekly search hours (winsorized). State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual; sample: employed workers. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Lagged expectations and timing. Appendix Table A7 replaces contemporaneous beliefs with beliefs reported in a prior core-survey wave that precedes the LMM interview. The persistence of the job-loss coefficient under lagged beliefs suggests that the belief–search relationship is not purely contemporaneous. While this does not fully resolve reverse-causality concerns, it indicates that beliefs measured before the search observation predict subsequent search intensity.

Two within-person specifications are estimated. First, a first-difference specification:

$$\Delta \text{SearchHours}_{it} = \beta' \Delta \mathbf{E}_{i,t-1} + \lambda_t + \Delta \varepsilon_{it}.$$

Second, an individual fixed-effects specification using lagged expectations:

$$\text{SearchHours}_{it} = \beta' \mathbf{E}_{i,t-1} + \alpha_i + \lambda_t + \varepsilon_{it},$$

where α_i denotes individual fixed effects. Standard errors are clustered at the individual level.

Within-person variation. Individual fixed-effects and first-difference specifications exploit within-person variation in beliefs over time. Statistical power is limited because most respondents appear only once or twice in the LMM. Consistent with this limitation, the within-person estimates are imprecise. They are nonetheless broadly consistent in sign with the cross-sectional evidence, particularly for voluntary separation beliefs.

B.3 Employed vs. Unemployed and Heterogeneity

The baseline specification is estimated separately for employed and unemployed workers. For the unemployed, the expectation vector includes macro pessimism and perceived job-finding probability (job-loss beliefs are not applicable):

$$\text{SearchHours}_{ist} = \beta' \mathbf{E}_{it} + \mathbf{X}'_{it} \gamma + \mu_s + \lambda_t + \varepsilon_{ist}.$$

Employment status comparison. Appendix Table A9 compares belief–search relationships across employment status. The belief–search relationship is strongest among employed workers, consistent with the interpretation that on-the-job search serves an insurance role against separation risk.

Table A8. Within-Person Identification: First Differences and Fixed Effects

	(1)	(2)	(3)	(4)
	FD Base	FD + Sep.	FE Base	FE + Sep.
$\Delta E_t[\Delta \Pr(U_{t+12} \geq 0)]$	1.335 (0.957)	0.905 (0.918)		
Lagged $E_t[\Delta \Pr(U_{t+12} \geq 0)]$			0.404 (0.543)	-0.201 (0.558)
$\Delta E_t \Pr(\text{Lose Job}_{t+12})$		1.334 (1.034)		
Lagged $E_t \Pr(\text{Lose Job}_{t+12})$				1.103 (0.834)
$\Delta E_t \Pr(\text{Leave Job}_{t+12})$		2.450*** (0.831)		
Lagged $E_t \Pr(\text{Leave Job}_{t+12})$				1.875*** (0.514)
Specification	First Diff	First Diff	Individual FE	Individual FE
Observations	1183	1111	4576	4402
R-squared	0.069	0.079	0.029	0.045

Notes: Columns 1–2 use first differences; columns 3–4 include individual fixed effects. Expectations scaled to $[0, 1]$. Time controls as in main specs; standard errors clustered by individual; short panel (average 1.4 LMM observations per worker) limits within-person power. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The baseline specification is estimated on subsamples defined by marital status, median income split, and age (< 40 vs. ≥ 40).

Worker heterogeneity. Appendix Table A10 explores heterogeneity by marital status, income, and age. The belief–search relationship is present across subgroups and is not confined to a narrow segment of the labor market.

B.4 Additional Validation of the Separation-Risk Channel

To further assess whether subjective job-loss beliefs contain information about realized labor-market risk, this subsection reports two additional validation exercises that are not used for identification in the main search regressions.

Prediction at multiple horizons. Appendix Table A11 re-estimates the validation specification at horizons from 1 to 9 months and with two right-censored outcomes for “any non-

Table A9. Expectations and Search: Employed vs. Unemployed Workers

	(1) Employed	(2) Unemployed
$E_t(\Delta \Pr(U_{t+12} \geq 0))$	0.953*** (0.355)	0.181 (1.318)
$E_t(\Pr(\text{Lose Job}_{t+12}))$	1.836*** (0.429)	
$E_t(\Pr(\text{Leave Job}_{t+12}))$	1.884*** (0.333)	
$E_t(\Pr(\text{Find} \text{Lose}_{t+3}))$	-1.156*** (0.294)	
$E_t(\Pr(\text{Find} \text{Unemp}_{t+3}))$		2.130** (1.018)
$E_t(\pi_{t+12})$ (dist. mean)	-0.028 (0.022)	0.134*** (0.039)
Sample	Employed	Unemployed
Observations	4130	712
R-squared	0.152	0.295

Notes: Employed sample: search hours on separation and job-finding expectations; unemployed sample: search hours on macro expectations and job-finding probability. Expectations scaled to $[0, 1]$. State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

employment within 12 months.” The coefficient on $P(\text{Lose Job})$ is positive and precisely estimated at short and medium horizons, and remains positive in the censored 12-month outcomes. This pattern indicates that the belief measure tracks realized separation-related risk over horizons that are relevant for employed search decisions.

Outcome-definition robustness. Appendix Table A12 compares two $t+3$ outcomes: broad non-employment/attachment and strict unemployment. The job-loss belief coefficient is positive and statistically significant in both columns. The precautionary interpretation therefore does not depend on one narrow coding of post-separation labor-market status.

Table A10. Heterogeneity by Marital Status, Income, and Age

	(1)	(2)	(3)	(4)	(5)	(6)
	Unmarried	Married	Low Inc.	High Inc.	Young	Old
$E_t[\Delta \Pr(U_{t+12} \geq 0)]$	1.125* (0.658)	0.887** (0.408)	1.254*** (0.467)	-0.030 (0.464)	1.272** (0.516)	0.647 (0.483)
$E_t \Pr(\text{Lose Job}_{t+12})$	2.525*** (0.672)	0.843 (0.552)	1.416** (0.572)	2.243*** (0.628)	1.587** (0.696)	1.906*** (0.578)
$E_t \Pr(\text{Leave Job}_{t+12})$	1.698*** (0.557)	2.311*** (0.419)	2.100*** (0.462)	1.723*** (0.442)	2.090*** (0.502)	1.985*** (0.473)
$E_t \Pr(\text{Find Job}_{t+3})$	-1.173*** (0.455)	-1.298*** (0.402)	-1.366*** (0.400)	-0.510 (0.401)	-0.862* (0.466)	-1.449*** (0.374)
Sample	Unmarried	Married	Low Income	High Income	Age < 40	Age ≥ 40
Observations	1675	2512	2844	1343	1989	2198
R-squared	0.221	0.189	0.164	0.139	0.187	0.207

Notes: Subsamples by marital status, income (median split), and age (< 40 vs ≥ 40). Expectations scaled to [0, 1]. State and time fixed effects with demographic controls; weighted by survey weights; standard errors clustered by individual. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11. Validation: Beliefs Predict Separations at Multiple Horizons (Q10-based)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	1m	3m	6m	9m	12m exact	Any<=12 (fup>=1)	Any<=12 (fup>=6)
$P(\text{Lose Job})$	0.0770*** (0.0065)	0.1290*** (0.0107)	0.1487*** (0.0156)	0.1266*** (0.0198)	0.5101 (0.3434)	0.1996*** (0.0159)	0.2504*** (0.0245)
Outcome	Not employed/attached	Not employed/attached	Not employed/attached	Not employed/attached	Not employed/attached (exact; selected)	Any not-employed (censored), fup≥1	Any not-employed (censored), fup≥6
Horizon	1 month	3 months	6 months	9 months	12 months	≤12 months	≤12 months
Observations	71116	55832	34656	16586	290	71116	27032
R-squared	0.171	0.205	0.287	0.415	0.690	0.206	0.349

Exact horizon outcomes require observation at t+h.

Any<=12 is right-censored and uses observed forward months; sample restrictions based on consecutive follow-up fup, consec12.

Table A12. Robustness: Beliefs Predict Both Non-Employment and Unemployment (Q10-based)

	(1)	(2)
	Not employed/attached (t+3)	Unemployed (t+3)
$P(\text{Lose Job})$	0.1290*** (0.0107)	0.0726*** (0.0079)
Observations	55832	55832
R-squared	0.205	0.186

B.5 2016 Election Event Study

This subsection provides an illustration of one source of within-location belief dispersion: partisan priors interacting with salient political events. Following [Mian et al. \(2018\)](#) and [Coibion et al. \(2020\)](#), the 2016 U.S. presidential election generated a sharp divergence in macroeconomic expectations across partisan-leaning groups. I use this episode to show that workers in similar local labor markets can hold systematically different macro views, and that these views can shift discontinuously around a major news event. I emphasize that this exercise is *illustrative*: it is not used as a stand-alone identification strategy for the causal effect of beliefs on search.

Design and sample. I classify states by whether the Republican or Democratic candidate won the state in November 2016. To focus on states with clearer partisan alignment, I drop swing states (defined based on the election margin threshold used in the replication code) and restrict attention to the 2016–2017 window. Because treatment is defined at the state level and the effective number of clusters is limited, results should be interpreted cautiously.

Event-study specification. I estimate an event-study that compares Republican-won and Democratic-won states around the election:

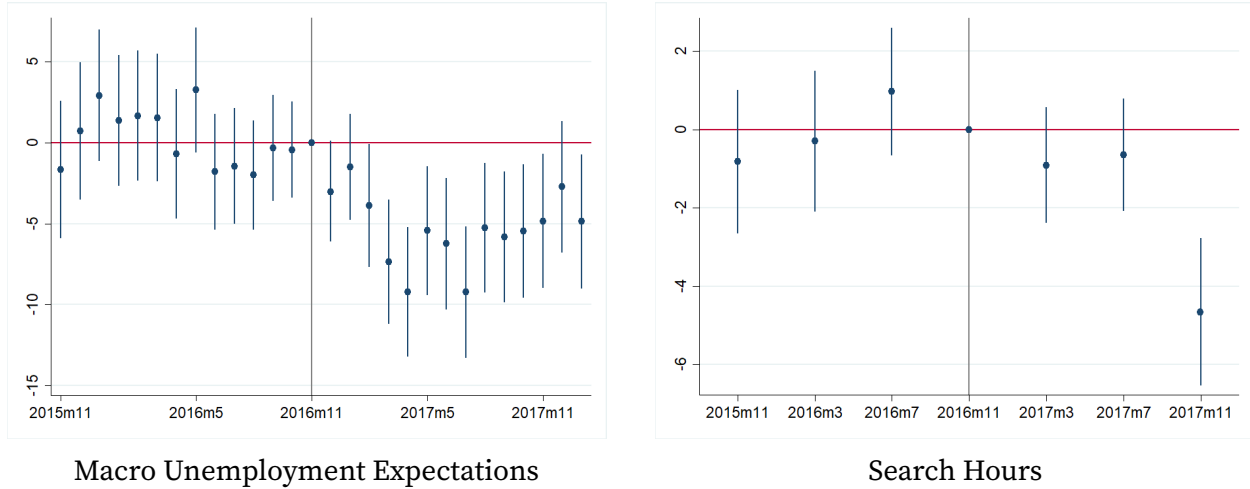
$$y_{ist} = \sum_{k \neq -1} \delta_k [\mathbb{1}\{t = k\} \times \text{RepWon}_s] + \mu_s + \lambda_t + \mathbf{X}'_{it} \gamma + \varepsilon_{ist}, \quad (30)$$

where t indexes event time (months relative to November 2016), μ_s and λ_t are state and month fixed effects, and $\mathbf{X}_{it}$ includes the baseline demographic controls. Standard errors are clustered at the state level.

Findings. Appendix Figure [A3](#) shows two patterns. First, macro unemployment expectations diverge sharply after the election: respondents in Republican-won states report a persistent decline in macro pessimism relative to respondents in Democratic-won states, with little evidence of differential pre-trends. This illustrates that belief dispersion about aggregate conditions can be substantial even among workers facing similar local environments.

Second, the corresponding series for search hours is consistent with lower post-election search in Democrat-won states relative to Republican-won states, but is imprecisely estimated. Precision is limited because search hours are observed only in the LMM (every four

months) and because the effective sample size is smaller once restricting to non-swing states. I therefore treat the search evidence as suggestive.



Notes: The figure plots event-study coefficients comparing Republican- and Democrat-won non-swing states around the November 2016 election. The left panel shows divergence in macro unemployment expectations; the right panel shows corresponding movements in search hours. Outcomes are weighted; specifications include state and month fixed effects. The quarterly LMM limits precision, so this evidence is illustrative rather than definitive.

Figure A3. Event Study: 2016 Presidential Election

The difference-in-differences specification is

$$y_{ist} = \delta (\text{RepWon}_s \times \text{Post}_t) + \mathbf{X}'_{it}\gamma + \mu_s + \lambda_t + \varepsilon_{ist},$$

where $\text{Post}_t = \mathbb{1}\{t \geq \text{Nov 2016}\}$ and the sample is restricted to non-swing states over 2016–2017. Standard errors are clustered at the state level.

The difference-in-differences specification confirms a sharp post-election divergence in macro unemployment expectations across Republican- and Democratic-won states. For search, the positive Rep-won $\times$ Post estimate implies lower post-election search in Democrat-won states relative to Republican-won states, but the estimate is imprecise, reflecting the low frequency of the LMM and the limited number of state-level clusters. I therefore interpret this exercise as an illustration of belief dispersion rather than a definitive estimate of the causal effect of macro beliefs on search.

Overall interpretation of robustness evidence. Across specifications, three patterns are robust:

1. The perceived job-loss probability consistently predicts higher search intensity.
2. The effect survives demanding geographic and sectoral fixed effects.

Table A13. 2016 Election (Appendix): DID (With Controls)

	(1)	(2)	(3)	(4)	(5)
	E(U exp)	P(Lose)	P(Leave)	Search (arc)	Search + beliefs
Rep-won $\times$ Post	-0.060*** (0.014)	0.014 (0.010)	0.034** (0.016)	0.142 (0.193)	0.162 (0.197)
Unemp exp					0.054 (0.172)
P9: P(Lose)					0.856*** (0.228)
<i>N</i>	20932	12256	12254	587	586

Notes: Difference-in-differences using Rep-won $\times$ Post (Nov 2016) for non-swing states, 2016–2017. Two-way fixed effects (state, month) with controls; standard errors clustered by state. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3. The relationship is present both cross-sectionally and, to the extent permitted by the short panel, in within-person specifications (which are imprecisely estimated due to limited LMM repeats).

These exercises do not establish causal identification of belief shocks. They instead demonstrate that the empirical relationship linking perceived job-loss risk to search behavior is not easily explained by local shocks, industry conditions, functional-form choices, or fixed individual heterogeneity.

Taken together with the validation evidence in Section 2.4 (beliefs predict subsequent separations) and the mechanism evidence in Section 2.5 (stated motives and interaction patterns), the robustness exercises strengthen the interpretation that employed workers adjust search effort in response to perceived separation risk.

C Model Appendix

Appendix Table A14 lists the parameter values used in the quantitative exercises. Parameters are monthly unless otherwise noted.

C.1 Some Model Derivations

This appendix collects short derivations that clarify (i) the mapping from hazards to monthly transition probabilities under the model’s within-month timing, (ii) the search first-order condition when the worker evaluates unemployment risk using *perceived* sep-

Table A14. Full Parameter List

Parameter	Description	Value
β	Discount factor	0.9967
ϕ	Worker bargaining weight	0.5
α	Matching elasticity (unemployment)	0.5
b	Flow value of unemployment	0.4
$y(G), y(B)$	Productivity levels	1.00, 0.95
Π_{GG}, Π_{BB}	Markov persistence (monthly)	0.99, 0.90
$\sigma(G), \sigma(B)$	Separation hazards (monthly)	0.024, 0.040
η	Employed/unemployed contact rate ratio	0.1
χ	Matching efficiency	0.338
κ	Vacancy posting cost	0.508
ψ	Search effectiveness	26.282
γ	Search cost curvature	0.0229
ζ	Belief level shifter	0.775

Notes: Parameters are monthly unless stated otherwise. The transition matrix has off-diagonal elements $\Pi_{GB} = 1 - \Pi_{GG}$ and $\Pi_{BG} = 1 - \Pi_{BB}$. Productivity is normalized to $y(G) \approx 1$. The baseline sets $m(G) = m(B) = 1$, so the cyclicalty of perceived risk is inherited from objective separation risk rather than belief distortions.

aration probabilities, and (iii) the benchmark Nash wage rule used in the quantitative implementation.

Throughout, time is monthly and the aggregate state follows $z \in \{G, B\}$ with transition matrix $\Pi(z, z')$.

C.2 Exact probability mapping and within-month timing

Let $\sigma(z)$ denote the *objective separation hazard* (per month) in state z . Under a Poisson arrival process within the month, the corresponding *monthly* separation probability is

$$\delta(z) \equiv 1 - \exp(-\sigma(z)), \quad \text{and survival probability } 1 - \delta(z) = \exp(-\sigma(z)). \quad (31)$$

Employed search effort s maps into an offer *hazard*

$$\xi(z, s) \equiv \lambda_e(\theta(z)) \psi g(s), \quad (32)$$

and therefore into an offer *probability*

$$p_{\text{offer}}(z, s) \equiv 1 - \exp(-\xi(z, s)). \quad (33)$$

Ordering within the month. The implementation follows the timing in Section 3.1: the current aggregate state z is realized, workers choose search, offers and separations are realized within the month in that same state, and then the economy transitions to z' for continuation values. With this ordering, the *objective* probability of ending the month unemployed, given current state z and effort s , is

$$\delta_{EU}(z, s) = (1 - p_{\text{offer}}(z, s)) \delta(z). \quad (34)$$

Equivalently, offers reduce unemployment inflows by reducing the probability that a separation “hits” a worker who lacks a fallback option.

C.3 Perceived separation risk and perceived employment-to-unemployment probability

Workers may perceive job-loss risk differently from the objective hazard. Let the *perceived* separation hazard be

$$\tilde{\sigma}(z) = \zeta \cdot m(z) \cdot \sigma(z), \quad (35)$$

where $\zeta > 0$ is a level shifter and $m(z)$ allows for state-dependent distortions.

Because the model uses hazard-consistent mapping, the perceived *monthly* separation probability is

$$\tilde{\delta}(z) \equiv 1 - \exp(-\tilde{\sigma}(z)). \quad (36)$$

Under the same within-month ordering (search and offer realization within month, with separation risk realized at month-end), the worker’s *perceived* probability of entering unemployment in current state z is

$$\tilde{\delta}_{EU}(z, s) = (1 - p_{\text{offer}}(z, s)) \tilde{\delta}(z). \quad (37)$$

This is the object that enters the worker continuation value in the implementation.

C.4 Deriving the search first-order condition

Fix current aggregate state z and let the worker choose effort $s(z) \equiv s$ after observing z . Let $W(z)$ and $U(z)$ denote the worker values, computed under the worker’s subjective transition probabilities only through $\tilde{\delta}_{EU}$.

The employed value satisfies

$$W(z) = w(z) - c(s) + \beta \sum_{z'} \Pi(z, z') \left[(1 - \tilde{\delta}_{EU}(z, s)) W(z') + \tilde{\delta}_{EU}(z, s) U(z') \right]. \quad (38)$$

Differentiate (38) with respect to s (holding $w(\cdot)$, $\theta(\cdot)$, and the continuation values $W(\cdot)$, $U(\cdot)$ fixed from the perspective of the individual worker's static choice):

$$\begin{aligned} 0 &= -c'(s) + \beta \sum_{z'} \Pi(z, z') \left[-\frac{\partial \tilde{\delta}_{EU}(z, s)}{\partial s} W(z') + \frac{\partial \tilde{\delta}_{EU}(z, s)}{\partial s} U(z') \right] \\ &= -c'(s) + \beta \left[-\frac{\partial \tilde{\delta}_{EU}(z, s)}{\partial s} \right] \sum_{z'} \Pi(z, z') (W(z') - U(z')). \end{aligned} \quad (39)$$

Using (37), and noting that $\tilde{\delta}(z)$ does not depend on s ,

$$\frac{\partial \tilde{\delta}_{EU}(z, s)}{\partial s} = -\tilde{\delta}(z) \frac{\partial p_{\text{offer}}(z, s)}{\partial s}. \quad (40)$$

Substitute (40) into (39) to obtain the search first-order condition:

$$c'(s(z)) = \beta \tilde{\delta}(z) \frac{\partial p_{\text{offer}}(z, s(z))}{\partial s} \sum_{z'} \Pi(z, z') (W(z') - U(z')). \quad (41)$$

Finally, with $p_{\text{offer}}(z, s) = 1 - \exp\{-\xi(z, s)\}$ and $\xi(z, s) = \lambda_e(\theta(z))\psi g(s)$,

$$\frac{\partial p_{\text{offer}}(z, s)}{\partial s} = \exp(-\xi(z, s)) \lambda_e(\theta(z)) \psi g'(s) \geq 0. \quad (42)$$

Economic interpretation. Equation (41) makes the mechanism transparent. Search is attractive when (i) perceived job-loss risk $\tilde{\delta}(z)$ is high, (ii) effort moves offer availability (high $\partial p_{\text{offer}}/\partial s$), and (iii) the expected continuation surplus, $\sum_{z'} \Pi(z, z') (W(z') - U(z'))$, is large.

C.5 Benchmark Nash wage with free entry

This subsection derives the benchmark wage rule used in the quantitative implementation. The key modeling choice is that wages are set as in a standard DMP environment and do not renegotiate within the month in response to the worker's search choice. Employed search affects flows (through (34)) but not the wage schedule.

Firm value of a filled job. Let $J(z)$ denote the firm's value of a filled job in state z , under the objective separation process. Because separations are realized within the current month in current state z , the firm Bellman equation is

$$J(z) = y(z) - w(z) + \beta \exp(-\sigma(z)) \sum_{z'} \Pi(z, z') J(z'). \quad (43)$$

Free entry. Let $V(z)$ be the value of a vacancy posted in state z . Free entry implies $V(z) = 0$. With within-month Poisson contact hazard $q(\theta(z)) = \chi \theta(z)^{-\alpha}$, the monthly filling probability is $p_f(z) = 1 - \exp\{-q(\theta(z))\}$. Posting a vacancy costs κ for the month; if the vacancy is filled, the firm obtains the continuation value of a filled job next month. The zero-profit condition is therefore

$$\kappa = \beta p_f(z) \sum_{z'} \Pi(z, z') J(z'), \quad \text{with } p_f(z) = 1 - \exp(-q(\theta(z))). \quad (44)$$

(If one uses the small-hazard approximation $p_f(z) \approx q(\theta(z))$, (44) reduces to $\kappa \approx \beta q(\theta(z)) \sum_{z'} \Pi(z, z') J(z')$.)

Nash bargaining. Let the worker and firm split the match surplus according to Nash bargaining with worker share $\phi \in (0, 1)$. Let the worker outside option be unemployment, with flow value b . Under the benchmark wage protocol, the worker's outside option is summarized by b and tightness; employed search affects unemployment risk but does not enter the bargaining problem directly.

Under standard DMP bargaining with free entry, the wage can be written in the familiar reduced form:

$$w(z) = (1 - \phi) b + \phi (y(z) + \kappa \theta(z)). \quad (45)$$

Connection to discrete-time exact probabilities. In discrete time, the term $\kappa \theta(z)$ corresponds to the expected hiring cost that the worker saves the firm by accepting a match. Under Poisson matching within the month, an exact discrete-time analogue replaces $\theta(z)$ by the ratio of worker and vacancy transition probabilities,

$$\kappa \theta(z) \approx \kappa \frac{p_u(z)}{p_f(z)}, \quad (46)$$

where $p_u(z) = 1 - \exp\{-\lambda_u(\theta(z))\}$ and $p_f(z) = 1 - \exp\{-q(\theta(z))\}$ with $\lambda_u(\theta) = \chi \theta^{1-\alpha}$ and $q(\theta) = \chi \theta^{-\alpha}$. For small within-month hazards, $p_u(z)/p_f(z) \approx \lambda_u(\theta(z))/q(\theta(z)) = \theta(z)$, so (45) is the standard continuous-time approximation.

C.6 Insurance-off algebraic comparison

The insurance channel is the difference between the unemployment inflow probability with and without offers being operative at separation time. Using (34) and the insurance-off counterpart $\delta_{EU}^{\text{off}}(z) = \delta(z)$, we have

$$\delta_{EU}^{\text{off}}(z) - \delta_{EU}(z, s) = \delta(z) p_{\text{offer}}(z, s). \quad (47)$$

Thus, the insurance value of search is mechanically larger when separations are more likely (high δ) and when offers are more likely to be in hand (high p_{offer}).

C.7 Proofs for Section 3.4

This appendix collects short proofs for the propositions in Section 3.4. The arguments rely on monotonicity in the search first-order condition (15) and on the mechanical mapping from offers to unemployment transitions (19).

Proof of Proposition 1. Fix z and let

$$S(z) \equiv \sum_{z'} \Pi(z, z') (W(z') - U(z')) > 0.$$

Define

$$F(s, \tilde{\delta}(z)) \equiv c'(s) - \beta \tilde{\delta}(z) p_{\text{offer},s}(z, s) S(z).$$

An interior optimum satisfies $F(s(z), \tilde{\delta}(z)) = 0$. By (18), $p_{\text{offer},s}(z, s) > 0$, so

$$F_{\tilde{\delta}} = -\beta p_{\text{offer},s}(z, s) S(z) < 0.$$

Also,

$$p_{\text{offer},ss}(z, s) = \exp\left(-\lambda_e(\theta(z))\psi g(s)\right) \lambda_e(\theta(z))\psi \left[g''(s) - \lambda_e(\theta(z))\psi (g'(s))^2\right] < 0,$$

because $g'(s) > 0$, $g''(s) \leq 0$, and $\lambda_e(\theta(z))\psi > 0$. Hence

$$F_s = c''(s) - \beta \tilde{\delta}(z) p_{\text{offer},ss}(z, s) S(z) > 0.$$

By the implicit function theorem,

$$\frac{\partial s(z)}{\partial \tilde{\delta}(z)} = -\frac{F_{\tilde{\delta}}}{F_s} > 0.$$

□

Proof of Proposition 2. From (19), $\delta_{EU}(z, s) = \delta(z)(1 - p_{\text{offer}}(z, s))$. Because $p_{\text{offer}}(z, s)$ is strictly increasing in s , $\delta_{EU}(z, s)$ is strictly decreasing in s . Because $\delta(z)$ is strictly increasing in $\sigma(z)$, $\delta_{EU}(z, s)$ is strictly increasing in $\sigma(z)$. □

Proof of Proposition 3. Define $F_z(s) \equiv c'(s) - \beta \mathcal{MB}(z; s)$. Under Assumption 1, $c'(s)$ is strictly increasing and, using (18), $\mathcal{MB}(z; s) = \tilde{\delta}(z) p_{\text{offer},s}(z, s) \sum_{z'} \Pi(z, z') (W(z') - U(z'))$. From the previous proof, $p_{\text{offer},ss}(z, s) < 0$, so

$$\frac{\partial \mathcal{MB}(z; s)}{\partial s} = \tilde{\delta}(z) p_{\text{offer},ss}(z, s) \sum_{z'} \Pi(z, z') (W(z') - U(z')) < 0.$$

Therefore,

$$F'_z(s) = c''(s) - \beta \frac{\partial \mathcal{MB}(z; s)}{\partial s} > 0,$$

so $F_z(s)$ is strictly increasing and has a unique root.

If $\mathcal{MB}(B; s) \geq \mathcal{MB}(G; s)$ in a neighborhood containing $\{s(G), s(B)\}$, then $F_B(s) \leq F_G(s)$ in that neighborhood. Since $F_G(s(G)) = 0$, it follows that $F_B(s(G)) \leq 0$. Because F_B is increasing and crosses zero at $s(B)$, we must have $s(B) \geq s(G)$. □

Proposition 5 (UI crowds out precautionary on-the-job search). *Holding fixed $\theta(\cdot)$, $\tilde{\sigma}(\cdot)$, and Π , a higher unemployment benefit b reduces the optimal on-the-job search intensity $s(z)$ for each aggregate state z .*

Proof of Proposition 5. Holding fixed $\theta(\cdot)$, $\tilde{\sigma}(\cdot)$, and Π , the only channel through which b affects the marginal benefit of search in (15) is the continuation surplus term $\sum_{z'} \Pi(z, z') (W(z') - U(z'))$. Under the maintained condition that $\partial(W(z') - U(z'))/\partial b < 0$ for each z' , this term decreases in b , so the right-hand side of (15) decreases in b . Since $c'(\cdot)$ is strictly increasing, the optimal choice $s(z)$ must decrease with b . The statement about unemployment inflows follows from Proposition 2. □

Proof of Proposition 4. If $s^{\text{base}}(B) > s^{\text{fix}}(B)$, then by Proposition 2, $\delta_{EU}(B, s^{\text{base}}(B)) < \delta_{EU}(B, s^{\text{fix}}(B))$. Holding $p_u(B)$ fixed, the flow-balance expression (16) is increasing in $\delta_{EU}(B, \cdot)$, so steady-state unemployment is lower under baseline search than under fixed search. □

C.8 Appendix: Insurance-Off Counterfactual

The following proposition formalizes the counterfactual discussed in Section 5.3. In the “insurance off” economy, offers do not prevent unemployment upon separation. Then the

perceived employment-to-unemployment probability simplifies to

$$\tilde{\delta}_{EU}^{\text{off}}(z) = \tilde{\delta}(z), \quad (48)$$

which is independent of search effort.

Proposition 6 (Insurance-off counterfactual). *Fix $z \in \{G, B\}$. If separated workers enter unemployment regardless of offer status, the employment-to-unemployment probability is*

$$\delta_{EU}^{\text{off}}(z) = \delta(z),$$

independent of search intensity.

Remark 2. In the baseline model,

$$\delta_{EU}^{\text{base}}(z, s) = \delta(z)(1 - p_{\text{offer}}(z, s)),$$

so search reduces unemployment inflows only through the offer channel. The gap between these expressions quantifies the insurance value of precautionary search emphasized in the counterfactual exercises.

This formal statement clarifies that the stabilizing effect identified in the main text operates entirely through the separation-to-unemployment margin. When that link is severed, search no longer affects unemployment inflows; in the baseline environment with no independent job-ladder or wage channel, the optimal precautionary search motive collapses.

C.9 Additional Counterfactual Checks

This subsection reports additional checks from the attempt7 counterfactual result files. These exercises are not separately targeted in calibration; they are reported to show that the mechanism is stable across related model runs.

The sign flip under constant separation risk confirms the main mechanism result: cyclical separation risk is the force that makes employed search countercyclical. The insurance-off benchmark keeps the search technology active but prevents offers from helping separated workers avoid unemployment. A full technology shutdown ($\psi = 0$) is not reported because it mechanically eliminates the return to search and moves the model to a corner.

Table A15. Additional Search-Cyclical Diagnostics

Scenario	$s(G)$	$s(B)$	$\Delta s = s(B) - s(G)$
Baseline (additional run)	0.566	0.716	+0.151
Constant separation risk	0.584	0.565	-0.020
Insurance off	0.559	0.703	+0.144

Notes: Values come from `exhibit1_search_cyclicalilty.txt` using the same `attempt7` calibration run (stage-0 + stage-2 mode-off). “Constant separation risk” sets $\sigma(G) = \sigma(B) = \bar{\sigma}$. “Insurance off” keeps offers active but prevents offers from covering separations. Positive Δs indicates countercyclical employed search.

Table A16. Marginal-Benefit Decomposition: Detailed Diagnostic

Counterfactual component held constant	$\Delta \mathcal{MB}$ (B-G)	Ratio to baseline
Baseline	0.002852	1.000
Perceived risk $\tilde{\sigma}$	-0.003234	-1.134
Offer response $\partial p_{\text{offer}}/\partial s$	0.003384	1.187
Continuation surplus $W - U$	0.003846	1.348

Notes: Values from `exerciseA_mb_decomposition.txt` under the same `attempt7` calibration run. $\Delta \mathcal{MB}$ is the recession-expansion difference in the marginal benefit index implied by the search FOC.

Holding perceived separation risk constant reverses the marginal-benefit differential, while holding offer response or continuation surplus constant strengthens it. This reinforces the economic ranking in the main text: cyclical risk is the dominant driver, and tightness/surplus movements are offsetting rather than reinforcing forces.

Table A17. Belief-Shock Dynamics with and without Insurance

Horizon (months)	Δs (on)	Δs (off)	$\Delta \delta_{EU}$ (on)	Δu (on, pp)
$t = 0$	0.0355	0.0332	-0.000431	-0.134
$t = 6$	0.0137	0.0128	-0.000168	-0.087
$t = 12$	0.0052	0.0049	-0.000064	-0.038
$t = 24$	0.0007	0.0007	-0.000009	-0.006

Notes: Values from `exhibit4_belief_shock_insurance.txt` under the same `attempt7` calibration run. The shock is a 10 pp increase in perceived job-loss risk in state B with persistence $\rho = 0.85$. At impact, offers reduce δ_{EU} by 36.29% relative to the insurance-off case.

Interpretation. When offers can prevent unemployment, the belief shock raises search and lowers unemployment inflows on impact and along the transition path. When offers cannot prevent unemployment, workers still search more, but unemployment inflows do

not fall. This supports the main text: search lowers unemployment in the model because offers help separated workers avoid unemployment.